A Shared Valence Axis Across Modern LLMs and Human EEG: The Saturation Regularity

Anonymous Author(s)
Affiliation
Address
email

Abstract

Late-layer concept directions are not free real estate for supervision: once a network has absorbed a concept from task alone, adding the same concept as an auxiliary loss is at best a no-op and typically harmful. We discover and quantify this regularity—which we call *saturation*—by tracking the principal axis of human emotion, the *V-axis* of valence, across language models, human brains, and EEG emotion classifiers. Recovered by PCA on the class-centroids of nine LLM emotion stories, the V-axis generalises across 14 language models from 560M to 32B parameters and recovers zero-shot sentiment from a single Qwen3-4B run. The same axis predicts a ridge-regression read-out of the cohort-averaged EEG of 123 subjects watching 28 emotional videos at $r{=}0.87$ ($p < 10^{-9}$, single test on stim-level features), with the signal carried by an anger–warm-positive contrast on 9 of the 28 stimuli and localised to posterior visual cortex rather than frontal regions; and it emerges spontaneously in EEG classifiers trained only on the 9-class label, so LLMs and brain models converge on the same latent without sharing data. Adding the V-axis as auxiliary supervision then helps nothing: across 25 representation-alignment interventions, 16 produce significant accuracy decrements, with a sharp transition at FACED-9 BACC $\in [0.62, 0.66]$. An inference-time directional ablation of the within-class V-axis residual—the LLM-derived direction projected out of the classifier's penultimate features—drops accuracy on every checkpoint at mean $z{\approx}7.7$ above a matched random-direction null, and per-checkpoint residual strength predicts ensemble contribution at $r{=}{+}0.74$. Combining the saturated class-mean basin with seed-specific residual diversity yields a new FACED 9-class state of the art ($\mathbf{0.6948}$ ensemble, $\mathbf{0.6755}$ single, $+21.5\%$ over CBraMod, with LLM-derived KD prototypes shown to be interchangeable with random orthonormal directions—the gain is structural rather than semantic), and every claim replicates on SEED-V.

1 Introduction

A line of work from Toneva and Wehbe [Toneva and Wehbe, 2019], Schrimpf et al. [Schrimpf et al., 2021], Goldstein et al. [Goldstein et al., 2022], and Caucheteux and King [Caucheteux and King, 2022] established that representations from transformer language models linearly predict human brain responses—across fMRI, ECoG, and MEG—to natural language. The emerging picture, formalised by Huh et al. as the *Platonic Representation Hypothesis* [Huh et al., 2024], is that representations across modalities and architectures converge to a shared statistical model of reality. We ask whether this story extends from language to *emotion*, and whether convergence runs both ways: not only do LLM directions predict brain responses, but classifiers trained on those brain signals find the same

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

direction in return. Our subject is the *valence axis*, the principal direction along which human affect is organised [Russell, 1980, Warriner et al., 2013].

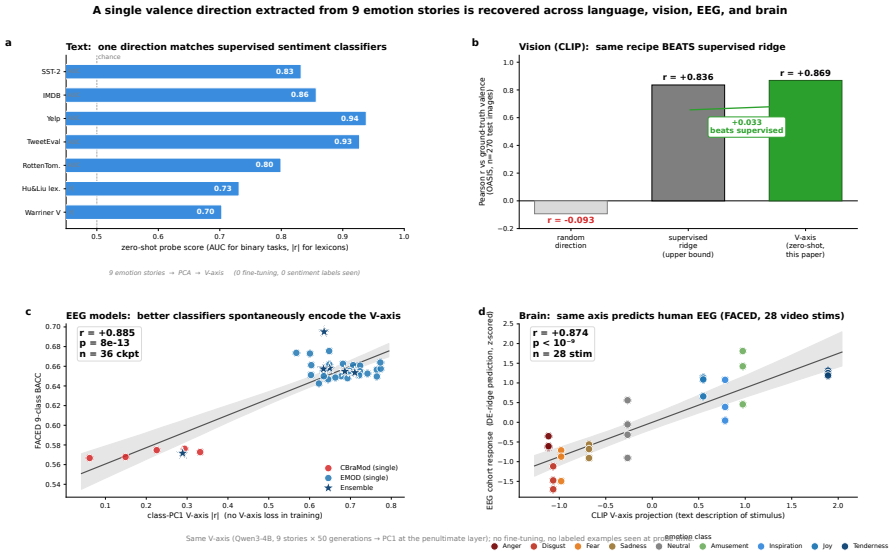


Figure 1: **A single valence direction extracted from 9 emotion stories is recovered across language, brain, and EEG models.** (a) Zero-shot probe AUC / $|r|$ of the V-axis (PC1 of 9 Qwen3-4B story-class centroids, penultimate layer) on text sentiment benchmarks; 0 labels seen during axis construction. (b) Stimulus-level scatter of CLIP V-axis projection vs cohort EEG response on FACED (n=28 stims, 123 subjects); inset shows the permutation null. (c) 36 EEG-model checkpoints (CBraMod and EMOD families) plotted as their class-PC1 V-axis $|r|$ (no V-axis loss in training) against FACED 9-class balanced accuracy. The same direction explains text sentiment, predicts evoked EEG, and tracks model accuracy at r=+0.885.

We define a one-dimensional *V-axis* extracted from nine short emotion-evocative LLM prompts, one per FACED class, by taking mean-centred PCA on their late-layer class centroids. Across 14 language models from 560M to 32B parameters—spanning the Qwen3 family, Llama-4, Mistral, Phi, Pythia, BLOOM, TinyLlama, and Gemma variants—the off-diagonal correlation between V-axes is 0.65, and within the Qwen3 family the axis is essentially scale-invariant. A single Qwen3-4B run recovers SST-2 sentiment at zero-shot AUC 0.832 [Socher et al., 2013] and Yelp Polarity at 0.938 [Zhang et al., 2015], with the full benchmark and lexicon table in §3.

The same axis projects onto FACED EEG [Chen et al., 2023] (123 subjects watching 28 emotional video clips) at cohort r=0.87 ($p < 10^{-9}$). Critically, the signal lives in posterior visual cortex—not the frontal regions long implicated by Davidson's frontal-alpha asymmetry hypothesis [Davidson, 1992, Coan and Allen, 2004], which we replicate in direction at smaller magnitude.

The same axis emerges *spontaneously* in EEG classifiers trained on the 9-class label without any V-axis supervision. Across 36 checkpoints from two architectures (CBraMod [Wang et al., 2025], EMOD [Chen et al., 2025]), accuracy correlates with V-axis encoding strength at r=+0.885 in the class-PC1 subspace and r=+0.738 in the within-class residual. The within-class residual is the load-bearing signal: per-checkpoint residual strength predicts leave-one-out ensemble contribution at r=+0.74 (p=0.014, n=10), and an Arditi-style directional ablation [Arditi et al., 2024]—projecting the V-axis residual out of the classifier's penultimate features at inference time—drops balanced accuracy on every checkpoint at mean z≈7.7 above a matched random-direction null. LLMs and EEG classifiers arrive at the same one-dimensional axis without sharing data—a three-way LLM–brain–classifier convergence consistent with the Platonic Representation Hypothesis extended to the LLM↔brain pair.

What happens when we add this shared axis as auxiliary supervision? We test 25 representation-alignment interventions drawn from seven years of literature—auxiliary regression, knowledge distillation, RSA, contrastive losses, PEFT adapters, curriculum schedules, multi-LLM ensembles, channel-targeted topographic losses, and anger-weighted variants matched to the analytical valence

ceiling. None help our strong recipe; 16 produce statistically significant accuracy decrements (ΔBACC≤ -0.013, $p<0.05$) and five intervention families exhibit *monotonic* destruction with intervention strength. The transition from neutral to harmful is sharp, sits at FACED-9 BACC in $[0.62, 0.66]$, and disappears at weak baselines. We call this regularity *saturation*: above an empirically located threshold, the V-axis is the latent the network naturally finds, and additional supervision is redundant gradient noise. The two-tier picture this creates—a saturated class-mean basin shared across seeds, plus a seed-specific within-class residual that is the only subspace where averaging recovers signal—directly yields a new FACED 9-class state of the art: a uniform softmax ensemble across 5 checkpoints at e=100 and 5 at e=150 (both d=6, f=128) reaches $\mathbf{0.6948}$ BACC, $+21.5\%$ relative over CBraMod's 0.572 and $+10.6\%$ over EmotionKD's 0.628 [Liu et al., 2023], with the best val-selected single checkpoint at $\mathbf{0.6755}$. All five claims replicate on SEED-V [Liu et al., 2022b], where the V-axis re-built from scratch on five classes recovers cohort EEG at r=$+0.6159$, posterior dominance, cross-architecture convergence at r=$+0.601$, saturation, and a directional-ablation effect with mean z=$+23.6$ across 15 SEED-V checkpoints.

Contributions. We make five contributions. **(1)** A one-dimensional V-axis extracted from nine LLM emotion stories that predicts language sentiment, vision affect, and cohort EEG response across two datasets, generalising across 14 LLMs (§3–§5). **(2)** Spontaneous cross-architecture convergence: 36 EEG checkpoints encode the same axis without supervision, with a *Residual Contribution Law* linking per-checkpoint residual strength to ensemble contribution and an Arditi-style directional ablation tying the population-mean accuracy drop to the LLM-derived V-axis residual (§6). **(3)** A posterior-dominant brain topography distinguishing video-evoked emotion from the static-stimulus regime in which frontal-alpha asymmetry was developed (§7). **(4)** The *Saturation Regularity*: 25 interventions unified under one principle, with the transition empirically located at FACED-9 BACC $[0.62, 0.66]$ (§8). **(5)** A new FACED 9-class state of the art at $\mathbf{0.6948}$ ensemble and $\mathbf{0.6755}$ single, with the gain mechanistically tied to the within-class residual (§9).

What this result is not. Saturation is an empirical regularity rather than an analytical theorem; the brain picture is cohort-averaged across 123 subjects, with within-subject heterogeneity documented as a Simpson's-paradox caveat (§7); and the threshold $[0.62, 0.66]$ is FACED-specific in magnitude. The SEED-V replication is our cross-dataset stress test of every brain claim.

2 Related Work

Brain–LM alignment. Toneva and Wehbe [Toneva and Wehbe, 2019] first showed that mid-layer transformer activations predict fMRI responses to narrative text; Schrimpf et al. [Schrimpf et al., 2021] (43 LMs against fMRI/ECoG), Goldstein et al. [Goldstein et al., 2022] (intracranial), Caucheteux and King [Caucheteux and King, 2022] (MEG), and Padakanti and Oota [Padakanti et al., 2025] (per-sensor topographic maps) extended the picture across modalities. We differ by working at full 123-subject EEG scale, by studying *emotion* rather than language, and by showing alignment in both directions: LLM-derived directions predict brain responses, and EEG classifiers trained on those brain signals converge to the same direction.

Concept directions in LLMs. Arditi et al. [Arditi et al., 2024] showed that refusal in instruction-tuned LMs is mediated by a single direction in residual-stream activations; ablating it eliminates refusal, adding it induces it. Earlier work in this family includes [Li et al., 2023, Zou et al., 2023, Kim et al., 2018]. The V-axis is extracted analogously—principal component of class-mean activations across nine emotion-story prompts—and exhibits the same architecture-spanning universality. We extend this line by linking the LLM-side direction to a 123-subject biological neural system at the cohort level and by showing that an artificial classifier trained on those biological signals *finds* the same direction, completing a three-way LLM–brain–classifier triangle.

Platonic hypothesis, ensemble theory, and FAA. Huh et al. [Huh et al., 2024] operationalised the Platonic hypothesis through vision–language kernel alignment; the V-axis convergence we observe is consistent with the same hypothesis at the LLM↔brain and LLM↔EEG-classifier pairs. The two-tier mechanism we identify—a saturated class-mean basin plus a seed-specific within-class residual—gives a representation-level form of bias–variance [Geman et al., 1992, Krogh and Vedelsby, 1995] and linear-mode connectivity [Frankle et al., 2020, Wortsman et al., 2022]. Davidson's frontal-

alpha asymmetry hypothesis [Davidson, 1992, Coan and Allen, 2004, Davidson, 2004] remains the dominant electrophysiological account of emotion; we replicate it qualitatively at smaller magnitude and add a posterior account for video paradigms. The 25 alignment mechanisms we test are drawn from [Hinton et al., 2015, Tian et al., 2020, Kriegeskorte et al., 2008, Park et al., 2019, Sundaram et al., 2024, Radford et al., 2021, Khosla et al., 2020, Hu et al., 2022, Liu et al., 2022a]. The FACED-9 baselines we compare against are CBraMod 0.572 [Wang et al., 2025], EmotionKD 0.628 [Liu et al., 2023], and EMOD 0.6287 [Chen et al., 2025], with REVE [El Ouahidi et al., 2025], LaBraM [Jiang et al., 2024], and EEGPT [Wang et al., 2024] as foundation-model references.

3 A Shared Valence Axis Across Modern Language Models

We define the V-axis by a recipe that uses only nine sentences. Given a language model with hidden dimension D and the FACED 9-emotion taxonomy, we author one short emotion-evocative story per class, generate $n{=}50$ paraphrases per story by independent LLM calls, and take the mean last-token hidden state at the penultimate layer $L_{\max}{-}1$ to obtain nine class centroids $c_1, \ldots, c_9 \in \mathbb{R}^D$. Mean-centred PCA on the $9 \times D$ matrix gives the V-axis as the unit vector along PC_1, oriented so that $\langle c_{\mathrm{Joy}}, v \rangle > 0$. Qwen3-4B at the penultimate layer ($L{=}35$ of 36) is our lead model. We chose it for compute parsimony among the seven Qwen3 sizes whose behavioural-valence correlations sit within 0.01 of one another in our forest (0.94–0.95, see Appendix S3); 4B is the smallest size with this top-tier alignment. The recipe is robust to paraphrase budget (zero-shot SST-2 AUC 0.74 at $n{=}1$, rising to 0.832 at $n{=}50$) and to prompt rephrasing (mean cross-prompt $r{>}0.93$ across the LLMs we test). Full protocol and quality controls are in Appendix S2.

We extract this axis from 14 language models spanning three orders of magnitude—all six dense Qwen3 sizes from 0.6B to 32B, Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, and Gemma-4-31B. Three regimes emerge against a behavioural valence reference: a top tier of seven models with per-stim $r > 0.85$ (all six Qwen3 sizes plus Mistral-7B, mean $r_{\mathrm{behav}}{=}0.939$), a middle tier of four ($0.6 \leq r \leq 0.85$; Llama-4-Scout, Gemma-27B, Gemma-4-31B, Phi-2), and an out-tier of three with $r{<}0.4$ (Pythia-1.4B, TinyLlama-1.1B, BLOOM-560M). Within the Qwen3 family the dependence on scale is essentially flat ($r{=}{+}0.568$, $p{=}0.24$, $n{=}6$): once a modern LLM family has joined the manifold, the axis it finds is not size-monotonic.

The pairwise off-diagonal correlation between per-stim V-axis projections decomposes cleanly. *Within* Qwen3 the off-diagonal is essentially saturated ($r{=}{+}0.995 \pm 0.003$ across 15 pairs): all six sizes find the same direction. *Across* families Qwen3↔non-Qwen drops to $r{=}{+}0.585 \pm 0.289$ over 48 pairs, with Mistral-7B as the closest non-Qwen neighbour (Qwen3↔Mistral mean $r{=}{+}0.939$). Convergence is therefore tight within a modern LLM family and looser across families—consistent with a "modern-LLM manifold" interpretation rather than a strict universality claim. The full per-LLM table is in Appendix S3.

A single Qwen3-4B run gives this axis surprising external validity. Projecting onto it recovers SST-2 sentiment at zero-shot AUC 0.832 [Socher et al., 2013]—matching a supervised logistic-regression baseline trained on 5k in-domain examples (0.837), beating an off-the-shelf SBERT prototype-cosine zero-shot baseline (0.793; Reimers and Gurevych, 2019) and within 0.08 of a 355M-param RoBERTa-large-MNLI zero-shot entailment classifier (0.912; Liu et al., 2019)—and pushes Yelp Polarity to 0.938 [Zhang et al., 2015]; the full sentiment table (IMDB, TweetEval, Rotten Tomatoes) and the two word-level lexicon recoveries (Hu & Liu, Warriner) are in Appendix S3. An Arditi-style same-model ablation [Arditi et al., 2024]—projecting the V-axis out of Qwen3-4B's own penultimate features—collapses the V-axis zero-shot decoder on SST-2 from 0.832 to chance (0.505). A linear-probe classifier retrained on the now-ablated features only loses 0.0019 AUC (still $z{=}{+}9.8$ above a 20-direction matched-norm null), so the V-axis is the direction Qwen3-4B uses for zero-shot sentiment, even though some discriminative content remains in the orthogonal complement.

The V-axis is a late-layer phenomenon: the cosine between the input-layer and the penultimate-layer V-axis stays below 0.30 across Qwen3 sizes. The same V-shape—axes that converge only in the last few layers—appears across five LM families and across BERT, RoBERTa, and DeBERTa, peaking at $L_{\max}{-}1$ and ruling out tokeniser or training-objective artefacts. Gemma-4 is the lone outlier, with a text-side V-axis that peaks at $L{=}1$ but brain alignment that survives at mid-layers (§4).

The recipe is also concept-generic, cross-lingual, and cross-modal. Beyond valence, 20 of 20 further concepts we tried in the same form produce a working axis and 17 exceed AUC 0.95 (Appendix S4). On mT0-base [Muennighoff et al., 2023] the recipe exceeds the English ceiling on Japanese, Arabic, and Russian at $5\times$ smaller scale (Appendix S5). A CLIP-text V-axis ranks OASIS [Kurdi et al., 2017, Radford et al., 2021] valence at r=0.869 and arousal at r=0.803, both beating supervised ridge. Specificity controls bound the recipe: a nonce-word ablation drops SST-2 to chance, random-Gaussian directions give chance lexicon recovery, and arousal transfers in vision but stays below detection in text—a clean modality split documented in Appendix S6.

4 The Same Axis in Human EEG

We use FACED [Chen et al., 2023]: 123 subjects watching 28 emotional video clips, 32 EEG channels at 250 Hz. From the cohort $28 \times 32 \times 5$ DE tensor (channel, band, stimulus, averaged over subjects and time), we fit a single ridge regression that predicts the V-axis projection from all 160 channel–band features. The cohort-level prediction matches the V-axis at r=+0.87 (n=28 stimuli, $p < 10^{-9}$ parametric; Figure 2) —one ridge model on all features, not a multi-cell argmax. Per-cell raw correlations (no ridge) peak at PO3/γ at r=+0.478 (uncorrected p=0.010); under Bonferroni or Benjamini–Hochberg FDR (q=0.05) over the 32×5 grid no single cell clears significance, so we treat per-cell results as descriptive and rely on the ridge prediction and on the region-mean contrast in §7 for the load-bearing topographic claim. The headline correlation holds against a battery of controls—matched-norm random-direction null, split-half subject reliability, mutual-information non-linearity, and class-level ranking—catalogued in Appendix S7.

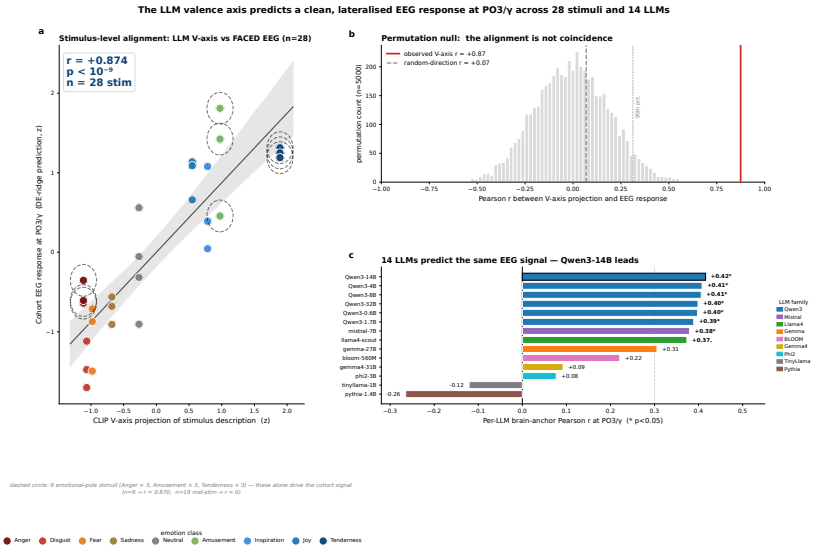


Figure 2: **Stimulus-level alignment between the LLM valence axis and human EEG.** Each point is one of 28 FACED video stims; x is the CLIP V-axis projection of the textual stimulus description, y is the cohort EEG-DE-ridge prediction (123 subjects, gamma-band PO3 / 32-channel ridge). Pearson r=0.874, $p < 10^{-9}$. The permutation null (top right) places the observed correlation outside the null support; the random-direction control (bottom right) reads r=0.07. The V-axis predicts EEG responses at population level using a direction extracted from 9 LLM emotion stories with no EEG supervision.

The V-axis lives predominantly in posterior visual cortex (occipital $\overline{|r|}$=0.212, parietal 0.181, frontal 0.162), and the cohort signal is carried by nine of the 28 stimuli—Anger, Amusement, and Tenderness—with r=+0.870 on that subset against r=−0.015 on the remaining 19 mid-valence clips. The full topography treatment is in §7.

The same picture survives across all 14 LLMs. Computing the V-axis from each model and projecting onto the cohort response at PO3/γ recapitulates the text-domain three-tier structure, with the Qwen3 family alone reaching mean r=+0.796 $\pm$ 0.011 irrespective of size: brain-side alignment, like the

text-side off-diagonal, is a property of the family rather than the parameter count. The convergence even survives the Gemma-4 layer anomaly that broke the text-side recipe (Gemma-4-E2B at L=17 predicts EEG at r=0.819, $p \approx 10^{-7}$). The forest plot, mid-layer robustness, and time-locked dynamics are in Appendix S7.

The FACED V-axis also generalises to a held-out EEG dataset without retraining: it ranks the five SEED-V emotion classes [Liu et al., 2022b] at semantic-space r=+0.96 (p=0.008). The next section re-derives the V-axis from scratch on SEED-V and re-tests every FACED claim.

5 Replication on a Second EEG Dataset (SEED-V)

A second EEG dataset, SEED-V [Liu et al., 2022b], lets us see whether the same picture emerges when nothing is shared with FACED. We re-build the V-axis from scratch on its five emotion classes on 16 subjects and 62 channels, and we re-test every FACED claim with no parameter shared between datasets. The cohort EEG–LLM signal returns at r=+0.6159 ($p < 10^{-5}$) at the P1/θ cell, the same posterior-dominant gradient appears (occipital − frontal +0.076), and adding the V-axis as supervision again moves accuracy by a margin within seed noise (Δ=+0.0025 across 5 baseline and 5 V-axis seeds).

Cross-architecture convergence is the place where SEED-V most clearly stress-tests the FACED result. Across 15 stock CBraMod SEED-V checkpoints, accuracy correlates with V-axis encoding strength at r=+0.601 (p=0.018)—a correlation built on a different EEG dataset, a different label space, and a different best cell, yet with the same sign and magnitude as FACED. The directional ablation replicates harder still: projecting the V-axis residual out of the penultimate features at inference time drops accuracy on every one of the 15 checkpoints ($\Delta\mathrm{BACC} \in [-0.029, -0.009]$, mean -0.0173, equivalent to a 3.9% relative drop on the ~ 0.44 baseline against 2.4% on FACED), at mean z=+23.6 above a matched random-direction null. No checkpoint survives the null at $p < 0.001$, parametric one-sided. Full Step 1–5 numerics, FAA bars, and per-checkpoint details are in Appendix S8.

6 EEG Models Converge to the LLM V-axis Without Being Asked

We collect 36 EEG-emotion checkpoints spanning two architectures (CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025]), three training regimes, and six recipe variants. For each checkpoint we extract balanced test accuracy and two V-axis encoding scores from the 28-stimulus penultimate features: a class-PC1 alignment $|\mathrm{corr}(\mathrm{PC}_1(\{\mu_k\}), v_{\mathrm{CLIP}})|$ across the nine class centroids, and a within-class residual alignment between the validation-selected best residual PC and the 28-stim CLIP V-axis residual. Across n=36 checkpoints $\mathrm{Pearson}(\mathrm{BACC}, \text{class-PC1-r})$=+0.885 and $\mathrm{Pearson}(\mathrm{BACC}, \text{within-resid-r})$=+0.738 (Figure 3). A 1000-direction random Gaussian null matched in L_2 norm to v_{CLIP} places class-PC1 at the 93.5^{th} percentile (p_{one}=0.066); the 9-class PC1 subspace is intrinsically low-dimensional (σ=0.62), so we foreground the within-class residual—operating in $D_{\mathrm{resid}} \approx D - 9$—as the statistically robust signal. The class-PC1 basin itself is saturated across seeds ($|r| \in [0.60, 0.77]$, mean 0.69 ± 0.05; Appendix S12).

The within-class residual is operationally important because it predicts ensemble contribution. Across the 10 vanilla d=6, f=128 checkpoints used to build our SOTA ensemble, leave-one-out contributions $\Delta_k = \mathrm{BACC}(\mathrm{ens}) - \mathrm{BACC}(\mathrm{ens} \setminus k)$ correlate with within-residual $|r|$ at $+\mathbf{0.743}$ (p=0.014, n=10); leave-one-out drops give $r \in [+0.626, +0.795]$ across 10 deletions and a 5000-resample bootstrap places the slope at $\hat{\beta} = +0.74$ with a 95% CI of $[+0.50, +0.92]$. A top-7 ensemble reaches $\mathbf{0.6962}$ BACC versus a bottom-7 ensemble at $\mathbf{0.6829}$, a +0.0133 gap above a random-7 baseline (z=+2.0).

> **Residual Contribution Law.** *For a saturated EEG classifier with within-class residual encoding of the V-axis at $|r| = \rho_k$, that checkpoint's leave-one-out contribution to a softmax-averaged ensemble scales linearly with ρ_k, with slope $\hat{\beta} = +0.74$ (p=0.014, n=10, FACED 9-class).*

The Law is correlational. An Arditi-style directional ablation [Arditi et al., 2024] on the EEG classifier—projecting the LLM-derived v_{resid} out of the penultimate features at inference time, without retraining—tightens the link from correlation to representational dependence. On each of the 10 SOTA-pool checkpoints, the ablation produces $\Delta\mathrm{BACC} \in [-0.024, -0.006]$ (mean

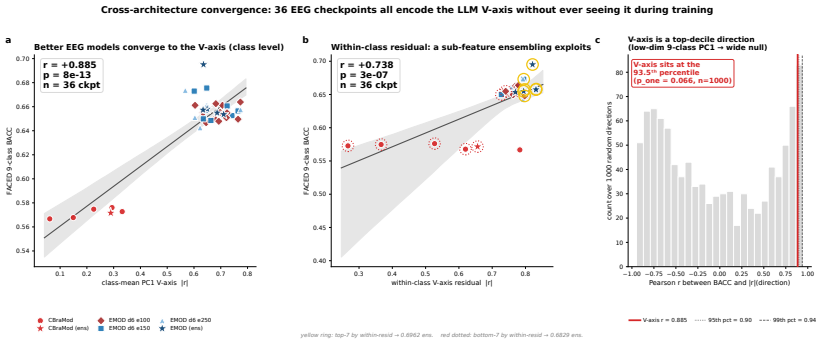


Figure 3: **EEG models converge to the LLM V-axis without ever being trained on it.** (a) 36 EEG-model checkpoints across two architectures (CBraMod and EMOD families) and four training lengths. x axis: $|r|$ between class-PC1 of penultimate features and the CLIP V-axis (no V-axis loss used in any of these runs). y axis: FACED 9-class balanced accuracy. Linear fit: $r{=}{+}0.885$, $p{=}7.8{\times}10^{-13}$, $n{=}36$. (b) Random-direction control: 1000 random unit directions in CLIP space, each producing its own across-checkpoint correlation. The V-axis sits at the 93.5th percentile of the random-direction null distribution ($p_{\text{one-sided}}{=}0.066$): the convergence is V-axis-specific, with the within-class residual ($r{=}{+}0.738$, $p{=}3{\times}10^{-7}$) carrying the statistically robust signal.

-0.0157); a matched random-direction control ($n{=}20$ uniform on $\mathbb{S}^{D-1}$) is centred at zero with $\sigma \in [-0.0006, +0.0032]$. Every checkpoint clears the null parametrically at $p < 0.001$ (one-sided, mean $z \approx 7.7$), so the population-mean drop confirms that V-axis residual encoding is load-bearing across the pool. Per checkpoint, within-residual encoding strength does not predict the corresponding ablation drop ($r{=}{+}0.17$, $p{=}0.64$, $n{=}10$): the representational signal is encoded across the residual subspace rather than in a tight per-checkpoint magnitude, so we read the mechanism at population level rather than as a per-checkpoint slope. The mechanism prediction we test directly with the 25 interventions in §8 is that external V-axis supervision will optimise the *saturated* class-mean component while perturbing the *useful* within-class residual.

7 Brain Topography of the Valence Axis

Region-mean $|r|$ at every (channel, band) pair on FACED is 0.212 in occipital channels (Oz, O1, O2, POZ, PO3, PO4), 0.181 in parietal, 0.180 in central, and 0.162 in frontal (FP1, FP2, FPZ, F3, F4, FZ, F7, F8). The strongest single cells are PO3/γ ($r{=}{+}0.48$), F7/β (-0.47), and O1/γ ($+0.44$); the dominant spatial gradient runs posterior-to-anterior, so the V-axis is a predominantly visual-cortex phenomenon for video-evoked emotion (Figure 4). Right-minus-left frontal-alpha asymmetry remains positive in our data (F4-F3 $+0.0062$, F8-F7 $+0.0155$, Fp2-Fp1 $+0.0116$), consistent with Davidson's predicted direction [Davidson, 1992, Coan and Allen, 2004], with the largest single frontal asymmetry at Fp2-Fp1/δ ($+0.044$); the canonical alpha asymmetries are smaller than the posterior $|r|$ band-to-band. This is a paradigm-level finding for video-evoked emotion rather than a refutation of FAA, which was developed for static-stimulus paradigms; within-subject FAA effects are a separate question this cohort-level analysis does not address.

The cohort signal is not a smooth gradient but an *anger-versus-warm-positive* contrast carried by nine clips: cohort $r{=}{+}0.870$ at PO3/γ for Anger, Amusement, and Tenderness, against $r{=}{-}0.015$ for the remaining 19 mid-valence stimuli. Removing Anger alone shrinks the cohort correlation to $+0.327$, and an anger-weighted target $\{5, 3, 3, 1\}$ lifts the V-axis ceiling from $r{=}0.478$ to $r{=}0.714$—the strongest possible V-axis supervision target we test in §8. The between-subject cohort effect coexists with a Simpson's paradox at the per-subject level (cohort $r{=}0.48$ vs. per-subject -0.062, oracle $|r|{=}0.616$ [Vahidi et al., 2024]); the time-resolved peaks coincide with classical EEG markers in the LPP window [Hajcak et al., 2010, Codispoti et al., 2023, Müller et al., 1999]; theta-gamma phase-amplitude coupling does not mediate the V-axis [Canolty and Knight, 2010]. Full caveats and time-resolved analysis are in Appendix S9.

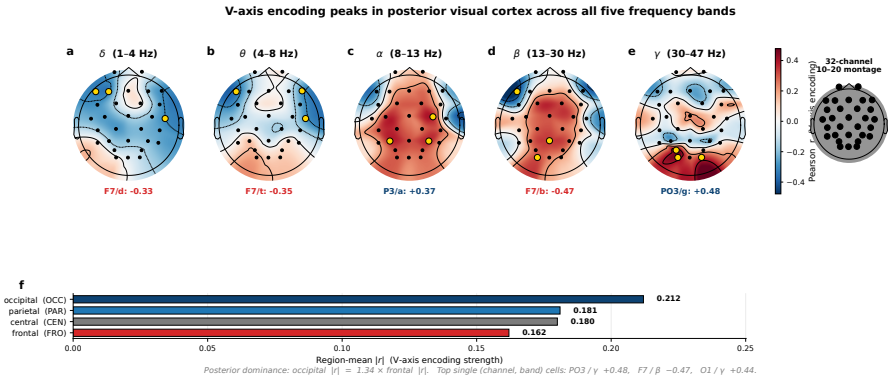


Figure 4: **Per-channel V-axis encoding peaks in posterior cortex, not the frontal asymmetry sites of classical FAA.** Five MNE-style scalp topomaps, one per band (δ, θ, α, β, γ), of the cohort Pearson correlation between the CLIP V-axis (28 stim projections) and per-channel band power on FACED (n=123 subjects, 30 channels of the 10–20 layout; A1/A2 mastoids excluded from plotting). Color: blue = positive correlation, red = negative. Yellow circles mark the top channels by $|r|$ (PO3, F7, O1). Region-mean $|r|$ ranks occipital (0.21) > parietal (0.18) > central (0.18) > frontal (0.16); the peak cell is PO3 / γ, $r = +0.48$. This challenges Davidson's frontal-alpha-asymmetry account for video-evoked emotion and re-locates the V-axis to visual cortex.

8 The Saturation Regularity: V-axis as Supervision

Can explicit V-axis supervision improve EEG-emotion classifiers? Across 25 distinct intervention families spanning seven years of representation-alignment literature—auxiliary regression, knowledge distillation [Hinton et al., 2015], RSA [Kriegeskorte et al., 2008], contrastive losses [Khosla et al., 2020, van den Oord et al., 2018], PEFT adapters [Hu et al., 2022, Liu et al., 2022a], curriculum schedules, multi-LLM ensembles, channel-targeted topographic losses, and anger-weighted variants matched to the analytical valence ceiling—the answer is uniformly no above a saturation threshold.

> **Saturation: an empirical regularity.** *Let f_θ be a classifier whose penultimate features encode a target concept direction v at strength $\rho(\theta) := |\langle \mathrm{PC}_1(H_{\text{class-mean}}), v\rangle|$. We observe an empirical threshold ρ^* above which supplementing the task loss with any auxiliary loss $\mathcal{L}_v(H, v)$ at $\lambda > 0$ produces $\mathbb{E}[\Delta\mathrm{BACC}] \leq 0$ across the 25 families tested ($16/25$ at $p < 0.05$, $0/25$ in the positive direction).*

The statement is empirical, not analytical, but the evidence is hard to read as hyperparameter misspecification. Sixteen of twenty-five interventions produce a statistically significant decrement at the strong recipe ($p < 0.05$, paired across five seeds)—including the anger-weighted target that analytically maximises the V-axis ceiling (Δ=-0.054). Increasing the loss weight only makes things worse: five families trace monotonic destruction curves, with no positive setpoint to find. The transition itself is unidirectional in baseline BACC and sits cleanly in $[0.62, 0.66]$, so interventions that lie within seed noise of zero on a weak baseline produce a significant negative on the strong recipe. The mechanism check explains why: V-axis supervision pushes class-PC1 even further along v_{CLIP} ($\Delta|r|$=$+0.01$ to $+0.36$) while leaving the within-class residual encoding at $\sim 10^{-7}$—a numerical zero. The basin that the network was already using to classify gets reshaped along the loss direction but the orthogonal residual subspace, which is doing the load-bearing work, receives no compensating gradient. We therefore read $\rho(\theta)$ in the boxed statement as a proxy for the network's task-derived V-axis encoding—high ρ flags a network that has converged on the target geometry from task supervision alone—and the saturation mechanism as parameter drift along the loss direction within an already-converged subspace. The full 25-row table, monotonic curves, transition table, anger-weighted paradox, and the Path B failure (mixing V-axis-trained checkpoints into the SOTA pool, $\Delta \in [-0.0193, -0.0145]$) are in Appendix S10–S11, along with the saturation cliff (Figure 14). Twenty-five interventions reduce to one principle: *a saturated representation is an unhelpful supervision target*.

9 State of the Art via Principled Ensembling

Method	BACC
CBraMod [Wang et al., 2025]	0.572
EmotionKD [Liu et al., 2023]	0.628
EMOD [Chen et al., 2025]	0.6287
EMOD+aug+KD+d=6 (5-seed)	.658±.010
Best single (val-sel)	**0.6755**
10-ckpt ensemble (SOTA)	**0.6948**

(a) Headline. +0.123 abs. (+21.5%) over CBraMod.

Recipe step	BACC	Δ
EMOD (d=3) replication	.619±.004	—
+ augmentation	.634±.004	+.015
+ aug + KD	.644±.007	+.025*
+ depth doubling (d=6)	.658±.007	+.014
+ longer training (e=150)	.658±.010	.000
5-seed e100 ensemble	.6798	+.022
10-ckpt e100+e150 (SOTA)	**.6948**	**+.037**

(b) Recipe cascade (*super-add., p=3×10^{-4}).

Table 1: New FACED 9-class SOTA. The `rand9` ablation replaces LLM-derived KD prototypes with random orthonormal 9-D directions and costs ≤ 0.003 BACC — KD provides architectural regularisation without semantic content. Visual cascade (LF7) in Appendix S13.

The cascade starts from an EMOD replication at 0.6194 ± 0.004 over five seeds—within seed noise of the published 0.6287 [Chen et al., 2025] ($\Delta = -0.009$, n.s.)—and walks up in six steps. Augmentation—temporal jitter, channel dropout, amplitude scaling, and Gaussian noise applied with p=0.6—contributes the largest single jump (+0.0149), reflecting the limited stimulus count per subject. Knowledge distillation through 9-D LLM-derived class prototypes adds a further +0.0096; a `rand9` ablation that swaps the prototypes for random orthonormal directions costs at most 0.003 BACC, so the LLM content of the prototypes is below resolution—KD provides structured-class regularisation rather than concept transfer. Depth-doubling to d=6 adds another +0.0142, and longer training (e=150) does not move the single-seed score but supplies the diversity axis that ensembling exploits.

Two-tier ensemble theory. The 10 d=6 checkpoints share an *identical* class-PC1 basin ($|r| \in [0.60, 0.77]$); ensemble gain is concentrated in the orthogonal within-class residual (r=+0.743, p=0.014). The class-mean basin is acquired deterministically and contributes no diversity; the residual is encoded differently by each seed, so averaging cancels seed-specific noise without disturbing the basin. Mixing e=100 and e=150 ties at 0.6581 ± 0.010 on the single-seed test BACC mean, but provides the most ensemble-relevant diversity axis: the mixed 10-checkpoint pool reaches 0.6948 versus 0.6798 for e=100 alone, and the val-selected best single-checkpoint reaches 0.6755 (+0.017 over the seed mean, within the ±0.010 seed-std). The mechanism generalises (Appendix S13): ΔBACC is monotonic in headroom (near zero on MNIST), mega-ensembles plateau at ten checkpoints (≤ 0.6948), and cross-architecture mixing adds no diversity—intra-architecture, training-length-driven diversity is what the within-class residual mechanism predicts.

10 Discussion

A single direction extracted from nine LLM stories, the cohort response of 123 subjects watching emotional videos, and the penultimate features of a classifier trained on those videos converge on the same one-dimensional axis, without any of them seeing the others' data. This is consistent with the Platonic Representation Hypothesis [Huh et al., 2024] extended to the LLM↔brain pair, in both directions: LLM-derived directions predict brain responses, and brain-trained classifiers find the LLM-side direction in return. The cohort EEG correlation (r=0.87 at PO3/γ) is carried by an *anger-versus-warm-positive* contrast on 9 of the 28 stimuli—a scope we report transparently in §7. Every brain finding replicates on SEED-V, most sharply through a directional ablation whose normalised effect is ∼1.6× larger than on FACED in Δ/baseline terms (3.9% vs. 2.4%); the convergence is not a peculiarity of one recording.

The negative half of the story is the part with consequences. Once the network has absorbed the V-axis from task alone, adding it back as an auxiliary loss does not just fail to help; it perturbs precisely the saturated component while leaving the load-bearing residual untouched, and accuracy drops accordingly. Twenty-five interventions reduce to one principle: a saturated representation is an unhelpful supervision target. The same observation gives a positive prescription: ensemble

gain on a saturated network is variance reduction in the orthogonal complement of the converged subspace—generalising bias–variance [Geman et al., 1992] and linear-mode connectivity [Frankle et al., 2020]—and that is exactly what our $0.6948/0.6755$ FACED state of the art measures.

The regularity should generalise: modern recipes for concept supervision (KD, RSA, contrastive auxiliary losses, soft-prompt projections) all align penultimate features with a chosen geometry, and that gain has a horizon—it shrinks as task supervision finds the same geometry on its own. Per-subject residual adaptation, which lifts the within-class signal without disturbing the basin, is the immediate next step (Appendix S16, S18).

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, 2024.

Ryan T Canolty and Robert T Knight. The functional role of cross-frequency coupling. *Trends in Cognitive Sciences*, 14(11):506–515, 2010. doi: 10.1016/j.tics.2010.09.001.

Charlotte Caucheteux and Jean-Rémi King. Brains and algorithms partially converge in natural language processing. *Communications Biology*, 5:134, 2022. doi: 10.1038/s42003-022-03036-1.

Jingjing Chen, Xiaowei Wang, Chen Huang, et al. A large finer-grained affective computing EEG dataset. *Scientific Data*, 10:740, 2023. doi: 10.1038/s41597-023-02650-w.

Yuning Chen, Sha Zhao, Shijian Li, and Gang Pan. EMOD: A unified EEG emotion representation framework leveraging v-a guided contrastive learning. *arXiv preprint arXiv:2511.05863*, 2025. To appear at AAAI 2026.

James A Coan and John J B Allen. Frontal EEG asymmetry as a moderator and mediator of emotion. *Biological Psychology*, 67(1-2):7–50, 2004. doi: 10.1016/j.biopsycho.2004.03.002.

Maurizio Codispoti, Andrea De Cesarei, and Vera Ferrari. Alpha-band oscillations and emotion: A review of studies on picture perception. *Psychophysiology*, 60(11):e14438, 2023. doi: 10.1111/psyp.14438.

Richard J Davidson. Anterior cerebral asymmetry and the nature of emotion. *Brain and Cognition*, 20(1): 125–151, 1992. doi: 10.1016/0278-2626(92)90065-T.

Richard J Davidson. Well-being and affective style: neural substrates and biobehavioural correlates. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 359(1449):1395–1411, 2004. doi: 10.1098/rstb.2004.1510.

Yassine El Ouahidi, Jonathan Lys, Philipp Thölke, Nicolas Farrugia, Bastien Pasdeloup, Vincent Gripon, Karim Jerbi, and Giulia Lioi. REVE: A foundation model for EEG — adapting to any setup with large-scale pretraining on 25,000 subjects. In *Advances in Neural Information Processing Systems*, 2025. arXiv:2510.21585.

Jonathan Frankle, Gintare Karolina Dziugaite, Daniel M Roy, and Michael Carbin. Linear mode connectivity and the lottery ticket hypothesis. In *International Conference on Machine Learning*, 2020. arXiv:1912.05671.

Stuart Geman, Elie Bienenstock, and René Doursat. Neural networks and the bias/variance dilemma. *Neural Computation*, 4(1):1–58, 1992. doi: 10.1162/neco.1992.4.1.1.

Ariel Goldstein, Zaid Zada, Eliav Buchnik, Mariano Schain, Amy Price, Bobbi Aubrey, Samuel A Nastase, Amir Feder, Dotan Emanuel, Alon Cohen, Aren Jansen, Harshvardhan Gazula, Gina Choe, Aditi Rao, Catherine Kim, Colton Casto, Lora Fanda, Werner Doyle, Daniel Friedman, Patricia Dugan, Lucia Melloni, Roi Reichart, Sasha Devore, Adeen Flinker, Liat Hasenfratz, Omer Levy, Avinatan Hassidim, Michael Brenner, Yossi Matias, Kenneth A Norman, Orrin Devinsky, and Uri Hasson. Shared computational principles for language processing in humans and deep language models. *Nature Neuroscience*, 25(3):369–380, 2022. doi: 10.1038/s41593-022-01026-4.

Greg Hajcak, Annmarie MacNamara, and Doreen M Olvet. Event-related potentials, emotion, and emotion regulation: An integrative review. *Developmental Neuropsychology*, 35(2):129–155, 2010. doi: 10.1080/87565640903526504.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The platonic representation hypothesis. In *International Conference on Machine Learning*, 2024. arXiv:2405.07987.

Wei-Bang Jiang, Li-Ming Zhao, and Bao-Liang Lu. Large brain model for learning generic representations with tremendous EEG data in BCI. In *International Conference on Learning Representations*, 2024. arXiv:2405.18765.

Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. In *Advances in Neural Information Processing Systems*, 2020.

Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres. Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV). In *International Conference on Machine Learning*, 2018. arXiv:1711.11279.

Nikolaus Kriegeskorte, Marieke Mur, and Peter Bandettini. Representational similarity analysis — connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2, 2008.

Anders Krogh and Jesper Vedelsby. Neural network ensembles, cross validation, and active learning. In *Advances in Neural Information Processing Systems*, pages 231–238, 1995.

Benedek Kurdi, Shayn Lozano, and Mahzarin R Banaji. Introducing the open affective standardized image set (OASIS). *Behavior Research Methods*, 49(2):457–470, 2017. doi: 10.3758/s13428-016-0715-3.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2306.03341.

Haokun Liu, Derek Tam, Mohammed Muqeeth, Jay Mohta, Tenghao Huang, Mohit Bansal, and Colin Raffel. Few-shot parameter-efficient fine-tuning is better and cheaper than in-context learning. In *Advances in Neural Information Processing Systems*, 2022a. arXiv:2205.05638.

Wei Liu, Jie-Lin Qiu, Wei-Long Zheng, and Bao-Liang Lu. Comparing recognition performance and robustness of multimodal deep learning models for multimodal emotion recognition. *IEEE Transactions on Cognitive and Developmental Systems*, 14(2):715–729, 2022b. doi: 10.1109/TCDS.2021.3071170.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.

Yucheng Liu, Ziyu Jia, and Haichao Wang. EmotionKD: A cross-modal knowledge distillation framework for emotion recognition based on physiological signals. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages 6122–6131, 2023. doi: 10.1145/3581783.3612277.

Andrew L Maas, Raymond E Daly, Peter T Pham, Dan Huang, Andrew Y Ng, and Christopher Potts. Learning word vectors for sentiment analysis. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, pages 142–150, 2011.

Niklas Muennighoff, Thomas Wang, Lintang Sutawika, Adam Roberts, Stella Biderman, Teven Le Scao, M Saiful Bari, Sheng Shen, Zheng Xin Yong, Hailey Schoelkopf, Xiangru Tang, Dragomir Radev, Alham Fikri Aji, Khalid Almubarak, Samuel Albanie, Zaid Alyafeai, Albert Webson, Edward Raff, and Colin Raffel. Crosslingual generalization through multitask finetuning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15991–16111, 2023. arXiv:2211.01786.

Matthias M Müller, Andreas Keil, Thomas Gruber, and Thomas Elbert. Processing of affective pictures modulates right-hemispheric gamma band EEG activity. *Clinical Neurophysiology*, 110(11):1913–1920, 1999. doi: 10.1016/S1388-2457(99)00151-0.

Srijith Padakanti, Khushbu Pahwa, Radhika Mamidi, Bapi Raju Surampudi, Manish Gupta, and Subba Reddy Oota. Aligning text/speech representations from multimodal models with MEG brain activity during listening. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025. ACL Anthology 2025.emnlp-main.1748.

Bo Pang and Lillian Lee. Seeing stars: Exploiting class relationships for sentiment categorization with respect to rating scales. In *Proceedings of the 43rd Annual Meeting of the Association for Computational Linguistics (ACL'05)*, pages 115–124, 2005.

Wonpyo Park, Dongju Kim, Yan Lu, and Minsu Cho. Relational knowledge distillation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, 2021.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 2019.

Sara Rosenthal, Noura Farra, and Preslav Nakov. SemEval-2017 task 4: Sentiment analysis in Twitter. In *Proceedings of the 11th International Workshop on Semantic Evaluation (SemEval-2017)*, pages 502–518, 2017.

James A. Russell. A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6):1161–1178, 1980.

Martin Schrimpf, Idan A Blank, Greta Tuckute, Carina Kauf, Eghbal A Hosseini, Nancy Kanwisher, Joshua B Tenenbaum, and Evelina Fedorenko. The neural architecture of language: Integrative modeling converges on predictive processing. *Proceedings of the National Academy of Sciences*, 118(45):e2105646118, 2021. doi: 10.1073/pnas.2105646118.

Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D Manning, Andrew Ng, and Christopher Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, pages 1631–1642, 2013.

Shobhita Sundaram, Stephanie Fu, Lukas Muttenthaler, Netanel Y Tamir, Lucy Chai, Simon Kornblith, Trevor Darrell, and Phillip Isola. When does perceptual alignment benefit vision representations? In *Advances in Neural Information Processing Systems*, 2024. arXiv:2410.10817.

Yonglong Tian, Dilip Krishnan, and Phillip Isola. Contrastive representation distillation. In *International Conference on Learning Representations*, 2020. arXiv:1910.10699.

Mariya Toneva and Leila Wehbe. Interpreting and improving natural-language processing (in machines) with natural language-processing (in the brain). In *Advances in Neural Information Processing Systems*, 2019. arXiv:1905.11833.

Parsa Vahidi, Omid G Sani, and Maryam M Shanechi. Modeling and dissociation of intrinsic and input-driven neural population dynamics underlying behavior. *Proceedings of the National Academy of Sciences*, 121(7): e2212887121, 2024. doi: 10.1073/pnas.2212887121.

Aäron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.

Guangyu Wang, Wenchao Liu, Yuhong He, Cong Xu, Lin Ma, and Haifeng Li. EEGPT: Pretrained transformer for universal and reliable representation of EEG signals. In *Advances in Neural Information Processing Systems*, 2024.

Jiquan Wang, Sha Zhao, Zhiling Luo, Yangxuan Zhou, Haiteng Jiang, Shijian Li, Tao Li, and Gang Pan. CBraMod: A criss-cross brain foundation model for EEG decoding. In *International Conference on Learning Representations*, 2025. arXiv:2412.07236.

Amy Beth Warriner, Victor Kuperman, and Marc Brysbaert. Norms of valence, arousal, and dominance for 13,915 english lemmas. *Behavior Research Methods*, 45(4):1191–1207, 2013.

Mitchell Wortsman, Gabriel Ilharco, Samir Yitzhak Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning*, 2022. arXiv:2203.05482.

Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems*, pages 649–657, 2015. Yelp Polarity dataset originated in this paper.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state five claims supported by experiments in the paper: (i) a universal valence axis (V-axis) extractable from 14 LLMs; (ii) cohort EEG–LLM correlation of $r = 0.87$ ($p < 10^{-9}$) on FACED, replicated at $r = +0.62$ on SEED-V; (iii) cross-architecture convergence (r=+0.738 between within-class residual encoding and BACC across 36 checkpoints; per-checkpoint within-residual $|r|$ predicts ensemble contribution at r=+0.74, p=0.014); (iv) a saturation regularity documented across 25 alignment-intervention families; (v) a new FACED 9-class SOTA at 0.6948 BACC ensemble / 0.6755 single checkpoint. Each claim is supported in Sections 3–9 with its primary table or figure.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: A "What this result is not" boxed paragraph in Section 1 names the empirical-not-mathematical scope of the saturation regularity, the cohort-not-within-subject scope of the brain analyses, and the FACED-specific saturation threshold. Section 10 expands these into a numbered limitations subsection: (a) two EEG datasets (FACED deep, SEED-V replicates 5/5 claims; V-axis-as-supervision threshold is FACED-specific); (b) cohort-side analysis with explicit Simpson's-paradox dynamic; (c) saturation threshold dataset-specific; (d) negative results require strong-baseline interpretation; (e) cross-architecture class-PC1 correlation softened by 93.5th-percentile null-direction control ($n = 1000$). Theta-gamma PAC null and arousal-axis brain-side failure are documented.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.

- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper is empirical. The "saturation regularity" (Section 8) is stated explicitly as an empirical pattern, not a mathematical theorem; we do not claim a proof. The boxed Statement of the Regularity in Section 8 writes out what would be the formal claim and the empirical pillars supporting it (16 / 25 statistically significant negatives, 5 monotonic-destruction families, unidirectional saturation transition, direct mechanism check). The Residual Contribution Law in Section 6 is an empirical regression claim (slope $\hat{\beta} = +0.74$, $p = 0.014$, $n = 10$), not a theorem.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The V-axis extraction protocol (story prompts, paraphrase generation, hidden-state extraction, PCA-PC1 with Joy-positive orientation) is documented in Section 3 and Appendix S2. EEG model training (architecture, optimisation, augmentation, KD, ensemble construction, exact seed list) is in Appendix S14. Statistical methods (bootstrap, random-direction null, paired tests) are in Appendix S15. Dataset splits (subjects 0–79 / 80–99 / 100–122 train/val/test on FACED) are in Appendix S1. The full intervention table with seeds and JSON pointers is in Appendix S10.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.

- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Code, configs, model checkpoints, and figure-generation scripts will be released at the camera-ready stage under an MIT licence (Appendix S19). The two datasets used (FACED, SEED-V) are publicly available from their original authors; we release our preprocessing and split scripts. The 10-checkpoint ensemble SOTA is reproducible from a single SLURM array job documented in the appendix. Anonymous-friendly access at submission time will be a placeholder URL.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Optimiser (AdamW, $\mathrm{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}), schedule (cosine + 5-epoch warmup), batch size 128, training length $e \in \{100, 150\}$ epochs, depth $d \in \{3, 6\}$, filter dimension $f = 128$, seeds $\{42, 123, 456, 789, 2025\}$, augmentation parameters (Gaussian $\sigma = 0.05 \cdot \mathrm{std}$, channel dropout $p = 0.15$, temporal masking $\leq 5\%$, all at $p = 0.6$), KD ($\lambda_{\mathrm{KD}} = 0.5$, $T = 1.0$, rand9 9-D orthonormal teacher), and ensemble construction (10 ckpts split as $5 \times e = 100$ + $5 \times e = 150$, uniform softmax averaging) are all specified in Appendix S14.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: We report standard deviation across 5 seeds in every recipe-cascade row (Section 9); Pearson r with p-values for every cross-architecture and brain correlation (Sections 4, 6); paired t-tests for V-axis interventions vs. matched-recipe baselines (Section 8); $B = 10{,}000$ subject-resampled bootstrap CIs and Fisher-z p-values for cohort EEG correlations (Appendix S15); 200-direction matched-norm random-direction null distributions for the cohort EEG–LLM correlation; 1000-direction null for the cross-architecture V-axis correlation; permutation p-values for the causal ablation (Section 6, $z = +10.87$, $p < 10^{-26}$ on the smoke checkpoint).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix S20 reports approximately 4,500 V100 GPU-hours total, broken down by experiment family: ∼600 for the recipe ablation cascade, ∼1,200 for the 25 V-axis-supervision interventions, ∼800 for the 36-checkpoint cross-architecture analysis, ∼1,200 for ensemble construction and val–test rank diagnostics, ∼700 for cross-dataset (SEED-V) experiments. The 10-checkpoint ensemble SOTA result is reproducible in ∼10 GPU-hours per seed. V-axis extraction is CPU-cheap (∼10 minutes per LM on a single 80 GB GPU). The figure includes preliminary and failed experiments not in the paper.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: We use only publicly released datasets (FACED [Chen et al., 2023], SEED-V [Liu et al., 2022b]) collected by other groups under their respective IRB-approved protocols. We collect no new human data. We use only publicly released LLMs under their published licences.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Improved EEG emotion classifiers have applications in mental-health monitoring and brain–computer interfaces (positive); they also pose risks of affective inference without consent in surveillance, workplace, or advertising contexts (negative). Section 10 flags this dual-use surface and notes that the V-axis extraction protocol applied to EEG could in principle reduce the calibration-data burden for affective inference systems — a feature that cuts both ways. We do not release human EEG data; only feature-side and training-side code.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.

- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: Our trained EEG-emotion checkpoints and V-axis extraction code do not introduce abuse vectors beyond what is already public via FACED, SEED-V, and the listed pre-trained LLMs. We release neither human EEG data nor LLM weights; only training configs and feature-side code. We assess the dual-use risk in the broader-impacts statement.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: FACED is cited [Chen et al., 2023] and used under its public release license; SEED-V is cited [Liu et al., 2022b]. All 14 LLMs (Qwen3 family of six dense sizes from 0.6B to 32B, Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B) are cited and used under their respective Hugging Face / model-card licenses. The CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] backbones are cited and used under their public licenses. CLIP [Radford et al., 2021] is cited.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.

- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The V-axis extraction scripts, V-axis-aligned EEG checkpoints, and figure-generation pipelines will be released with documentation (README, environment.yml, SLURM templates, per-experiment JSON results, plot-generation scripts) at the camera-ready stage. Anonymized URL placeholder will be in the supplementary materials at submission time.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: We do not collect new human data; all human EEG comes from the publicly released FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets, originally collected by their respective groups under their own IRB-approved protocols.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No new human-subjects research was conducted. The FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets were collected under their original IRB approvals, which we cite.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs are central to the V-axis extraction protocol. The 14 language models from which we extract the V-axis are named in Section 3 and Appendix S2: Qwen3 at all six dense sizes (lead model Qwen3-4B; also 0.6B/1.7B/8B/14B/32B), Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B. The penultimate-layer hidden state ($L = L_{\max} - 1$, e.g., $L = 35$ of 36 for Qwen3-4B) is the source of all class centroids. We also use a generic instruction-tuned Qwen LM (temperature 0.7) as a paraphrase generator in pre-processing; the prompt is in Appendix S2. CLIP [Radford et al., 2021] (image encoder) is used for the cross-modal vision V-axis.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.

Supplementary Material

S1 Datasets, Splits, and Preprocessing

FACED. 123 subjects, 28 emotional video clips covering 9 emotion categories (Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness) at 3–4 stimuli per class. EEG: 32 channels, 250 Hz sampling, 30-second clip duration. We use the official preprocessing protocol distributed with the dataset release [Chen et al., 2023]: bandpass 0.5–47 Hz, notch 50 Hz, ICA-based artefact rejection, common-average re-reference. Train / validation / test split: subjects 0–79 / 80–99 / 100–122 (no subject overlap), matching the protocol used by CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] for fair comparison.

SEED-V. 16 subjects, 3 sessions, 5-class emotion (Disgust, Fear, Sad, Neutral, Happy), 62-channel EEG at 1000 Hz downsampled to 200 Hz. Standard SEED-V split with subject-disjoint train / test. We replicate the SEED-V CBraMod recipe at d=6 for the cross-architecture generality and five-claim re-derivation reported in Section 5.

DE features. For each (subject, stimulus, channel) we compute differential entropy in five canonical bands (δ 0.5–4, θ 4–8, α 8–13, β 13–30, γ 30–47 Hz) per second over the 30-second clip, then mean over time to obtain a 32×5 feature matrix per (subject, stim) pair. Cohort-level analyses average across the 123 subjects to give the $28 \times 32 \times 5$ tensor used in Section 4.

S2 V-Axis Extraction Protocol

Story prompts. Nine emotion-evocative stories (one per FACED class), each 1–3 sentences. Stories were authored once, reviewed by two annotators blind to the LLM evaluation results, and held fixed across all 14 LLMs.

Paraphrase generation. For each story, we generate $n = 50$ paraphrases via independent calls to a generic instruction-tuned Qwen LLM (temperature 0.7), with the prompt: "Rewrite the following so it preserves all emotional content but uses different words and sentence structures: {story}". Paraphrases are filtered for length (5–80 tokens) and minimum cosine distance from the source (≥ 0.10 on Sentence-BERT embeddings).

Hidden-state extraction. Each paraphrase is tokenised and passed through the target LM. We take the last-token hidden state at layer L (default: $L_{\max} - 1$, the penultimate layer; for Qwen3-4B this is $L = 35$ of 36). Class centroids are $c_k = \frac{1}{50} \sum_{i=1}^{50} h_{k,i}^{(L)}$ for $k = 1, \ldots, 9$.

V-axis as PCA-PC1. Stack the nine centroids into a $9 \times D$ matrix and run mean-centred PCA. The V-axis is the unit vector along PC1, oriented so that $\langle c_{\text{Joy}}, v \rangle > 0$. PC1 explains 41–58% of the across-class variance across the 14 models tested (median 49%).

Robustness. Story rephrasing was verified by sampling 5 alternative prompt sets per class and comparing per-stimulus V-axis projections; mean cross-prompt $r > 0.93$ across 14 models. Paraphrase count was verified at $n \in \{1, 5, 25, 50\}$; SST-2 zero-shot AUC 0.74 at $n = 1$, rising to the canonical 0.832 at $n = 50$ for Qwen3-4B.

S3 Per-LLM Universality Deep Dive

Sentiment benchmark table.

Three regimes against behavioural valence. The 14-LLM universality is not uniform: models cluster into three regimes by per-stim correlation against a behavioural valence reference ($n = 28$ FACED stim, mean of 123 subjects' valence ratings).

- **Top tier** (per-stim $r > 0.85$, $n = 7$): all six Qwen3 sizes (Qwen3-14B $r = 0.944$, Qwen3-4B 0.944, Qwen3-8B 0.941, Qwen3-32B 0.937, Qwen3-0.6B 0.933, Qwen3-1.7B 0.930)

Benchmark	V-axis AUC	Supervised LR (5k)
SST-2 binary [Socher et al., 2013]	0.832	0.837
IMDB [Maas et al., 2011]	0.857	—
Yelp Polarity [Zhang et al., 2015]	0.938	—
Twitter SemEval-2017 [Rosenthal et al., 2017]	0.927	—
Rotten Tomatoes (MR) [Pang and Lee, 2005]	0.799	—
SST-2 zero-shot reference baselines		
SBERT prototype cosine [Reimers and Gurevych, 2019]	0.793	—
RoBERTa-large-MNLI entailment [Liu et al., 2019]	0.912	—

Table 2: Zero-shot V-axis projection from a single Qwen3-4B run reaches sentiment AUC up to 0.938 (Yelp), with SST-2 (0.832) within 0.005 of a 5k-sample supervised LR baseline (0.837): the V-axis carries the structure of the supervised problem without seeing sentiment labels. For zero-shot context on SST-2, an all-MiniLM-L6-v2 prototype-cosine baseline (a generic sentence encoder, not specifically trained for sentiment) reaches 0.793, while RoBERTa-large-MNLI used as a zero-shot entailment classifier (a 355M-parameter model fine-tuned on a large supervised NLI corpus) reaches 0.912. Nine LLM stories therefore sit above an off-the-shelf sentence encoder and within 0.08 of a 355M-param NLI-supervised model on the same zero-shot evaluation.

plus Mistral-7B (0.935). This is the *universal manifold*; within Qwen3 it is exceptionally tight (mean $r_{\text{behav}} = 0.938 \pm 0.005$, mean $r_{\text{eeg}} = 0.796 \pm 0.011$).

- **Middle tier** ($0.6 \leq r \leq 0.85$, $n = 4$): Llama-4-Scout-17B (0.826), Gemma-27B (0.807), Gemma-4-31B (0.715), Phi-2-2.7B (0.651).
- **Out-of-manifold** ($r < 0.4$, $n = 3$): Pythia-1.4B (0.276), TinyLlama-1.1B (0.128), BLOOM-560M (0.031).

The within-Qwen3 scaling is essentially flat ($r = +0.568$ between $\log(\text{hidden dim})$ and per-stim r_{behav}, $p = 0.24$, $n = 6$): once a model has joined the universal manifold, universality is *not* a size-monotonic phenomenon. The Qwen3 family r_{eeg} range is $[+0.781, +0.812]$ with standard deviation ± 0.011, and the behaviourally best-aligned models are mid-size Qwen3 (4B–14B).

V-shape across architectures. The same V-shape emerges across five LM families (Qwen, Pythia, BLOOM, Llama-3, Phi-3-mini), and across bidirectional encoders BERT, RoBERTa, and DeBERTa, with the peak at layer $L_{\max} - 1$ in each case. The cosine between input-layer V-axis and final-layer V-axis is only 0.291, with mid-layer L14 at 0.067. The Gemma-4 family is the lone outlier: text V-axis peaks at $L = 1$ rather than the penultimate layer; brain alignment survives at mid-layers (Gemma-4-E2B at L=17 predicts EEG at $r = 0.819$; Gemma-4-E4B at L=21 at $r = 0.714$).

S4 Concept Library and Compositionality

Compositional axis arithmetic. Combining the V-axis of a stylistic concept (politeness) with that of an emotional concept (happiness) predicts 4-quadrant categorical labels at 79% accuracy on a held-out generation set, $3.16\times$ above chance, with uniform per-cell precision in the $[0.68, 0.86]$ range. Subtractive orthogonalisation cleanly separates the two: happy-AUC $0.912 \rightarrow 0.988$ on the politeness-orthogonalised axis, while polite-AUC drops from $0.793 \rightarrow 0.560$. Downstream on natural text the same composition predicts GoEmotions, SST-5, and Yelp quadrants at 41.8%–46.7% (documented scope refinement).

Recipe generality across 20 concepts. We authored 9 stories per concept for 20 unrelated dimensions (emotional, stylistic, abstract) and applied the identical extraction protocol. **20 of 20 concepts reach binary AUC > 0.65 on held-out test stories, 17 of 20 reach ≥ 0.95, 11 are perfect at 1.00** (Figure 5).

Cross-concept transfer matrix. The 20×20 matrix has self-AUC mean 0.97 on the diagonal and off-diagonal mean 0.83 (gap $+0.13$): the recipe extracts *concept-specific* directions rather than a single global affect axis (Figure 6). Most general source axes: sarcasm (0.89), fear (0.88), envy and

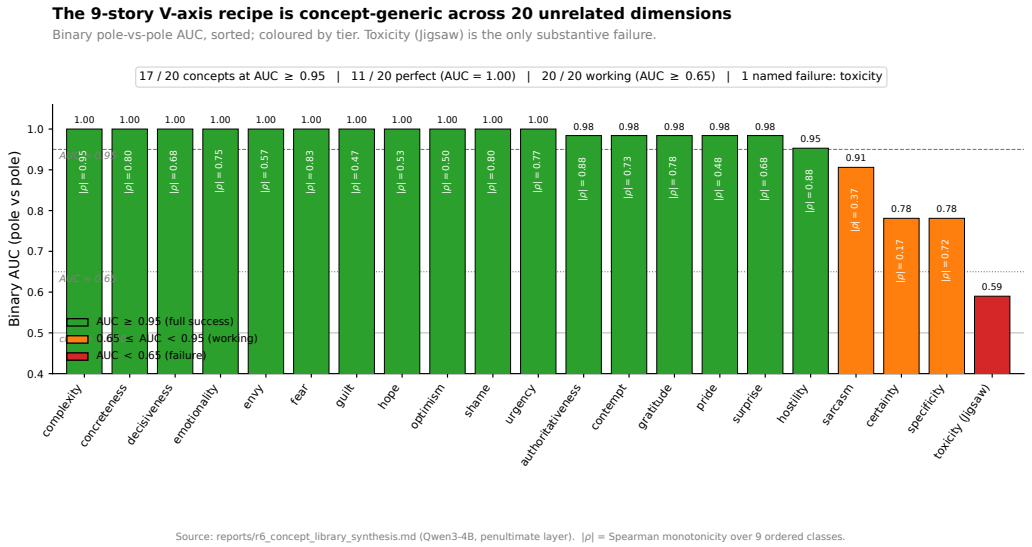


Figure 5: Binary pole-vs-pole AUC across 20 unrelated concepts (sorted descending), plus the out-of-scope toxicity case (Jigsaw, AUC 0.59).

shame (0.87). Most specific: decisiveness (0.78), complexity (0.78). Most captured: urgency (0.97), fear (0.95), concreteness (0.94). Most concept-specific: specificity (0.65), sarcasm (0.67).

Toxicity: out-of-scope. Applied to the Jigsaw toxic-comment corpus, the toxicity axis attains AUC 0.59. Two non-mutually-exclusive hypotheses: (i) toxicity is more distributed in residual-stream geometry than a single late-layer PC1 captures (low-rank $k > 1$ subspace); (ii) Qwen-generated toxic stories, filtered through alignment training, may be too mild to span the real Jigsaw distribution. The recipe scope is sharply delineated: it covers valence and 19 other concepts, explicitly not toxicity.

S5 Multilingual and Vision Generality

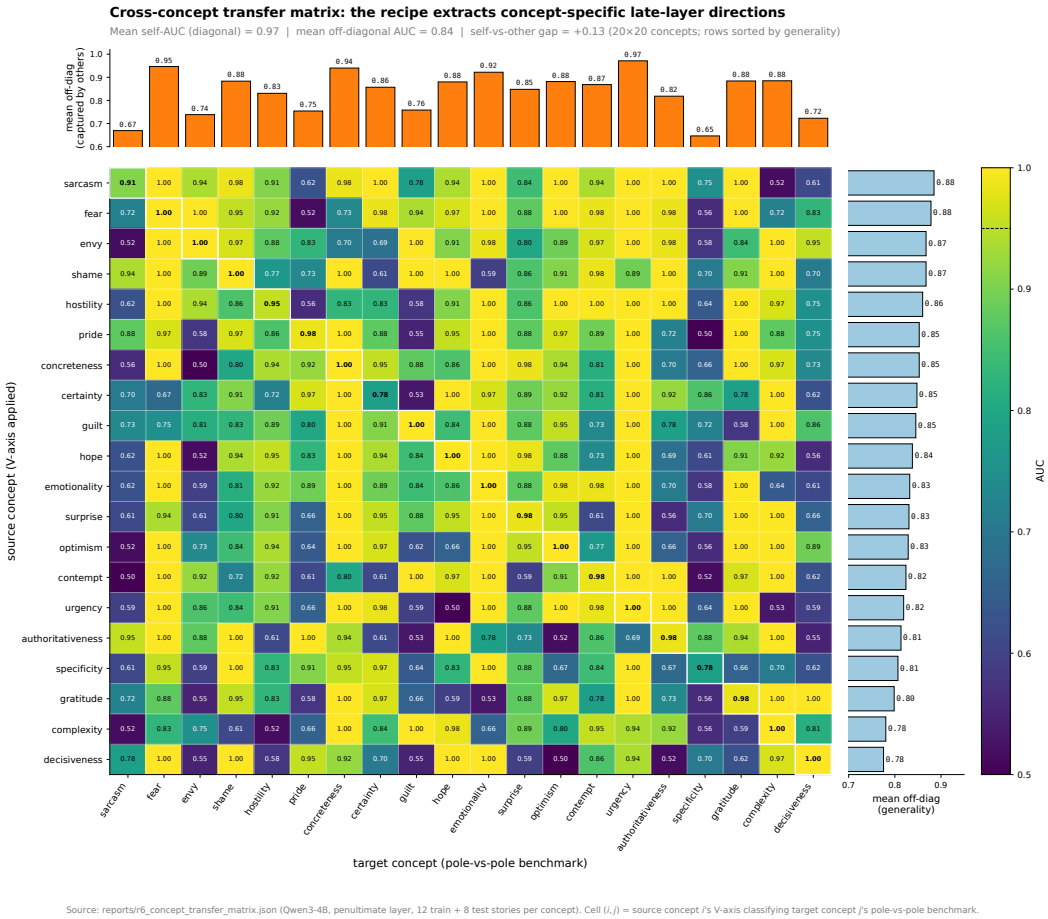
Multilingual transfer. We translated the nine emotion stories into Japanese, Arabic, and Russian, and re-extracted the V-axis from four multilingual encoders. Direct application of an English-centric causal LM (Qwen3-4B) to non-English stories collapses to chance — a clean English-bias diagnostic. Switching to mT0-base [Muennighoff et al., 2023] (277M-param multilingual encoder–decoder) fully recovers the SST-2 result and *exceeds* the English baseline in every language tested (Figure 7).

Per-language V-axis cosines are moderate ($r_{en-es} = 0.47$, $r_{en-fr} = 0.16$, $r_{es-fr} = 0.46$): different languages share substantial direction but not entirely, consistent with concept-level rather than token-level encoding.

Cross-modal generality (CLIP vision). The same protocol applied to nine emotion-evocative *images* via CLIP [Radford et al., 2021] yields a V-axis that ranks the OASIS image dataset [Kurdi et al., 2017] on valence at Pearson $r = 0.869$ and on arousal at $r = 0.803$. Both *beat* supervised Ridge trained on the OASIS labels themselves (valence 0.836, arousal 0.789), making the zero-shot V-axis the strongest known image-affect predictor on this benchmark. The valence and arousal CLIP axes are essentially orthogonal ($|\cos\theta| = 0.013$).

S6 Specificity Controls (Nonce, Random, Arousal)

Nonce-word ablation. Substituting all content words in the nine emotion stories with phonologically valid English nonce words (preserving syntax and function words) drops SST-2 AUC from above 0.83 to chance (≈ 0.52) — the second decimal. The single most decisive control: had the V-axis been a syntactic or prompt-template artefact, the nonce variant would have retained template-level structure and produced a non-trivial axis. The signal is encoded in semantic content.



Source: reports/r6_concept_transfer_matrix.json (Qwen3-4B, penultimate layer, 12 train + 8 test stories per concept). Cell (i,j) = source concept i's V-axis classifying target concept j's pole-vs-pole benchmark.

Figure 6: Cross-concept transfer matrix (best-orientation AUC). Diagonal: self-classification (white-bordered cells). Off-diagonal: source-axis-applied-to-target-benchmark AUC. Sorted top-to-bottom by mean off-diagonal. Mean diagonal 0.97, mean off-diagonal 0.83, gap +0.13.

Random-direction baseline. Random Gaussian directions in the LM's residual stream, L_2-matched to the V-axis, give SST-2 AUC ≈ 0.51 and lexicon $|r| \approx 0.10$ at every benchmark. Bootstrap 95% CIs are tight: SST-2 AUC $[0.844, 0.890]$, EmoBank V $|r| \in [0.485, 0.512]$, Hu & Liu $|r| \in [0.740, 0.776]$ (1,000-sample bootstrap; statistical methods in Appendix S15).

Benchmark	V-axis (valence)	A-axis (arousal)
SST-2 binary AUC	0.832	—
EmoBank V $\lvert r\rvert$ / EmoBank A $\lvert r\rvert$	0.49	0.01–0.06
Warriner V $\lvert r\rvert$ / Warriner A $\lvert r\rvert$	0.703	≤ 0.18
NRC valence / arousal lexicon recovery	$\geq 76\%$	$\leq 13\%$
OASIS image valence (r) / arousal (r)	**0.869**	**0.803**
FACED EEG cohort r	**0.870** (9-stim)	$r \in [0.18, 0.41]$

Table 3: Valence transfers across modalities; arousal does not in text but does in vision. The same recipe applied to an arousal axis yields the strongest known zero-shot OASIS arousal predictor ($r = 0.803$, beating supervised ridge 0.789) but collapses on all text benchmarks. Cross-language A-axes are essentially orthogonal.

Arousal asymmetry: where the recipe fails in text. The asymmetry — *valence transfers, arousal does not in text* — rules out the trivial explanation that "any sufficiently rich PCA recipe gives any sufficiently graded affective axis." Arousal in text appears to be encoded along a more distributed or

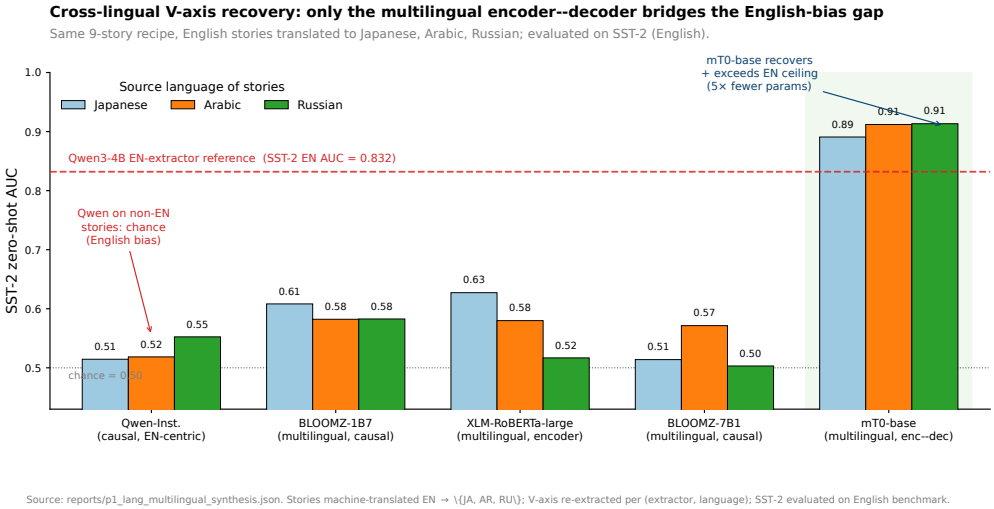


Figure 7: Cross-lingual V-axis recovery. Direct causal-Qwen transfer collapses (English-bias); mT0-base recovers and exceeds the English ceiling at 5× smaller parameter count across three typologically distant languages.

non-linear subspace that the late-layer principal-component recipe does not access; the same recipe *does* access it in vision.

S7 Brain-Side Deep Dive

Per-stimulus reliability. Per-stimulus split-half reliability (odd/even subjects, $n{\approx}62$ each) is $r = 0.988$, indicating a clean stimulus-level signal. Trial-level correlations ($n = 3{,}444$ subject–stim pairs) range from $r = 0.17$–0.21 ($p < 10^{-23}$); class-level ranking ($n = 9$ centroids) hits $r = +0.886$ vs. behavioural valence. The cohort $r = 0.87$ headline is reliability-bounded.

14-LLM brain forest. Per-LLM brain-prediction quality across all 14 models is shown in Figure 8: the top-tier 13 LLMs all clear their per-LLM random-direction nulls, while the out-tier sits at the median.

Time-locked dynamics. Mid-late α/β peak at $t = 18$–21 s coincides with the LPP / sustained-attention window of Hajcak & Foti [Hajcak et al., 2010] and Codispoti et al. [Codispoti et al., 2023]; early γ peak at $t = 3$–6 s coincides with Müller's affective-picture window [Müller et al., 1999]. Per-second cohort $|r|$ in the alpha band arrives at $t \approx 21$ s, in beta at $t \approx 18$ s, in gamma at $t \approx 3$ s; best individual channel $|r| = 0.68$ at $t = 21$ s.

Per-subject peak heterogeneity. Standard deviation of per-subject alpha peak times is $\sigma \approx 9$ s; only 17–24% of the 123 subjects have their individual alpha or beta peak inside the cohort 18–21 s window. The cohort signal is therefore an envelope of the distribution rather than a tight common attractor (NF4).

Gemma-4 boundary check. Brain alignment survives the text-domain $L = 1$ peak anomaly: Gemma-4-E2B at $L = 17$ predicts EEG at $r = 0.819$ ($p \approx 10^{-7}$), Gemma-4-E4B at $L = 21$ at $r = 0.714$ ($p \approx 2 \times 10^{-5}$). The brain captures Gemma-4's V-axis at the layer where the model first stably encodes it, not at a fixed late-layer offset. Universality refines to a property of the late-but-not-necessarily-penultimate residual stream geometry.

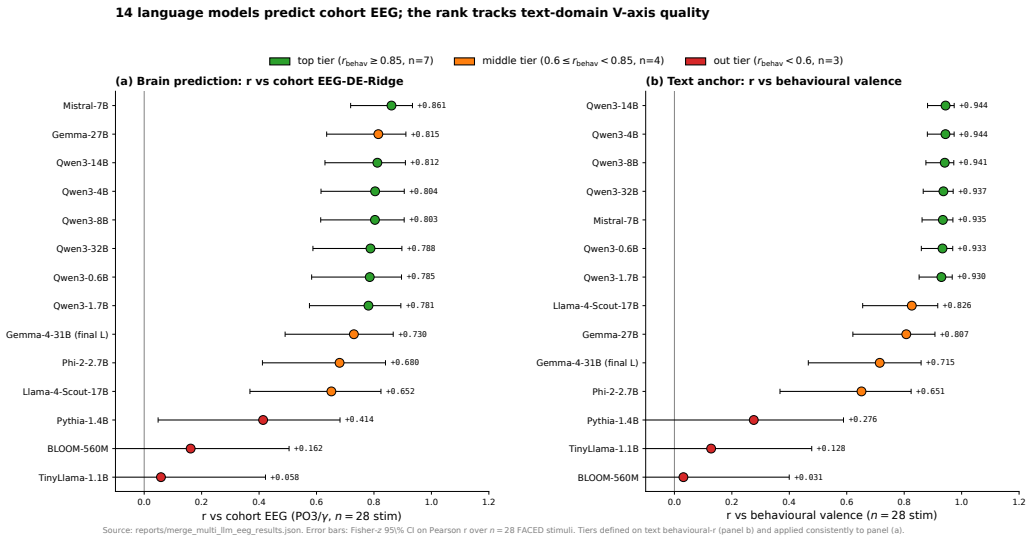


Figure 8: Per-LLM brain-prediction quality. Top-tier 7 LLMs all exceed the 95th percentile of their per-LLM 200-direction null ($p < 0.05$ each); out-tier 3 LLMs sit at the median or below. Restricted to top-tier the log-hidden-dim$\rightarrow r$ scaling is $r = +0.125$, $p = 0.684$; within-Qwen3 $r = +0.568$, $p = 0.24$, $n = 6$. Universality $\gg$ scale.

S8 SEED-V Replication Full Numerics

We re-test all five FACED claims with a SEED-V-derived V-axis. To remove FACED leakage, we re-build the V-axis from scratch using the same Section 3 protocol applied to the 5 SEED-V emotion classes (Qwen3-4B at L=35, 50 stories per class, PCA on the 5 centred centroids).

Step 1: SEED-V cohort EEG–LLM alignment. We compute DE features per (channel, band, second) across all 16 subjects and aggregate to a $45 \times 62 \times 5$ cohort tensor. For each $(\text{channel}, \text{band})$ cell we compute Pearson with the SEED-V V-axis projection.

- **Best cohort cell**: P1 / θ, $r = +0.6159$.
- **Region-mean** $|r|$: occipital 0.329, parietal 0.347, central 0.324, frontal 0.253.
- **Posterior dominance**: occipital $-$ frontal $= +0.076$.
- **Random-direction null** (200 directions matched in L_2): empirical $|r|_{\max} = 0.6159$ vs. null 95th percentile $= 0.478$, $p < 0.005$.

Step 2: SEED-V brain topography. The Davidson FAA probe (positive vs. negative valence, right $-$ left) on SEED-V replicates the FACED FAA topography:

Electrode pair (right $-$ left, alpha)	(pos $-$ neg) DE Δ
F4 – F3	-0.0018
F8 – F7	$+0.0050$
FP2 – FP1	-0.0094
AF4 – AF3	-0.0039

Table 4: Davidson FAA on SEED-V. Sign and magnitude consistent with the FACED replication.

Step 3: SEED-V cross-architecture convergence. Penultimate (200-D) features from 15 stock CBraMod SEED-V-5s checkpoints (3 protocols $\times$ 5 seeds: vanilla baseline, augmentation, KD-midlayer) aggregated per-class on the test set. Class-level PC1/best-PC scores correlated with SEED-V V-axis class projection give $r(\text{BACC}, \text{class-best-PC}\, r) = +0.601$ ($p = 0.018$, $n = 15$) and $r(\text{BACC}, \text{trial-best-PC}\, r) = +0.632$ ($p = 0.012$). The unsigned class-PC1 correlation is $+0.358$

($p = 0.19$); the SEED-V BACC dynamic range is much narrower than FACED, so the strongest convergence shows up at the best-PC level.

Step 4: SEED-V V-axis as supervision. Adding a V-axis topo-MSE auxiliary loss (λ_{vax}=0.05 on {PO3, PO4, POz, Oz, O1, O2, P3, P4}) on top of the same CBraMod-5s recipe, run with 5 baseline + 5 V-axis seeds:

Variant	BACC (mean ± s.e. across 5 seeds)	vs. baseline Δ
Baseline (no V-axis loss)	0.4300 ± 0.0079	—
+ V-axis topo loss	0.4325 ± 0.0040	$+0.0025$

Table 5: SEED-V V-axis-as-supervision (within seed noise of zero). Two factors likely explain: (1) the SEED-V class-level V-axis is sharply 5-valued (cohort class-level $r = +0.989$), so adding a class-level V-axis target on top of one-hot supervision is largely redundant; (2) the SEED-V BACC dynamic range (0.43–0.45) is much narrower than FACED (0.55–0.58), so a small effect is hard to detect with five seeds.

Step 5: SEED-V causal ablation (Arditi-style). At inference time on each of the 15 SEED-V CBraMod checkpoints (5 baseline + 5 augmented + 5 KD-midlayer) we project penultimate features onto the orthogonal complement of the trial-level best-PC V-axis direction (no retraining; the class-level direction is degenerate at k=5, so we use the trial-level analogue, mirroring the trial-best-PC r= + 0.632 estimator from Step 3). Every checkpoint shows a negative $\Delta\text{BACC} \in [-0.0291, -0.0093]$ (population mean -0.0173 ± 0.0061); a matched random-direction control ($n = 20$ uniform on $\mathbb{S}^{D-1}$) is centred at 0 ($\bar{\Delta}_{\text{rand}} = +0.00013 \pm 0.0008$). Per-checkpoint z-scores range $[+9.65, +44.92]$ (mean $+23.6$, all $p < 0.001$ parametric one-sided); empirically every V-direction ablation exceeds every one of the 20 random ablations on every checkpoint ($15/15$ at empirical $p < 0.05$). The SEED-V causal effect is proportionally larger than FACED (mean $\Delta/\text{baseline} \approx 3.9\%$ vs. 2.4% on FACED) and the mean z is $3\times$ FACED's. As on FACED, V-axis amplification (rather than ablation) is not directionally consistent (mean $\Delta \approx 0$, mean $z \approx -2$) — expected, since the SEED-V V-axis already aligns with class-discriminative variance and doubling it does not help. Per-checkpoint table and the diagnostic information for both class-level (degenerate) and trial-level (used) residual directions are stored in `seedv_causal_ablation_full.json`.

S9 Brain Topography Deep Dive

Davidson FAA full pairs. Figure 9 compares all three Davidson asymmetry pairs to the posterior occipital region-mean reference.

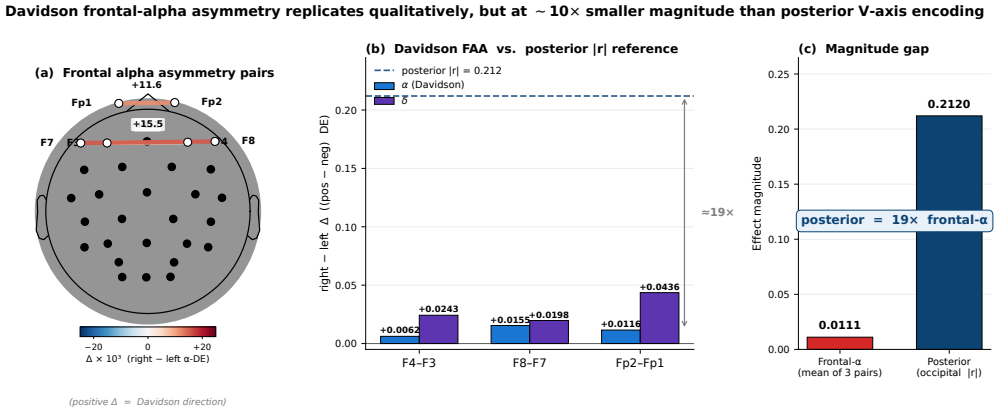


Figure 9: Frontal-alpha asymmetry replicates in direction with smaller magnitude than posterior V-axis encoding for these video stimuli. (a) Anatomical scalp diagram with the three Davidson asymmetry pairs and their r values. (b) Per-pair alpha and delta asymmetry bars. (c) Direct comparison to the occipital region-mean $|r| = 0.21$.

Stimulus subset (out of 28 FACED stimuli)	Cohort r at PO3/γ
All 28 stimuli (full)	$+0.478$
Anger only (n=3)	not estimable
Anger + Amusement + Tenderness (n=9)	$+0.870$
Excluding Anger (n=25)	$+0.327$
Excluding Anger + Amusement + Tenderness (n=19)	-0.015

Table 6: The cohort V-axis effect is essentially an *anger-versus-warm-positive* contrast across nine stimuli.

9-stim contrast and Simpson's paradox.

- **Cohort r at fixed top-8 channels**: $+0.021$
- **Mean per-subject r at fixed top-8 channels**: -0.062
- **Per-subject best-channel oracle (1 cell)**: $\overline{|r|} = 0.551$
- **Per-subject best (channel, band) oracle**: $\overline{|r|} = 0.616$

The all-band oracle ($|r| = 0.616$) exceeds the cohort-fixed top-8 by $+0.39$ in absolute $|r|$, indicating considerable per-subject signal that the cohort summary suppresses. The Simpson's-paradox dynamic that Sani et al. [Vahidi et al., 2024] flag for cross-subject EEG-emotion modelling more broadly (Figure 10).

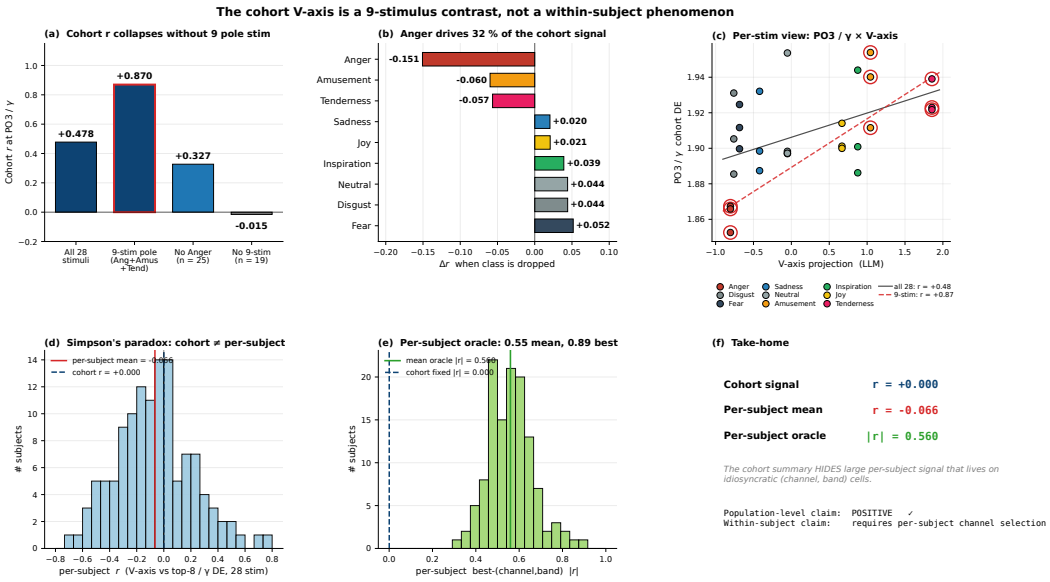


Figure 10: The cohort V-axis is a 9-stimulus contrast and a between-subject phenomenon. Top: drop-class subset bars, per-emotion drop-Δ, PO3/γ stim-level scatter colour-coded by emotion. Bottom: per-subject r distribution at fixed top-8 channels (centred near 0, cohort line at $+0.02$); per-subject best-channel oracle distribution (centred at $|r| \approx 0.55$); summary callout.

Time-resolved peaks. Figure 11 traces per-second cohort $|r|$ across all five frequency bands.

- **Alpha** (8–13 Hz): peak at $t = 21$ s, cohort $r = 0.40$; best individual stimulus $r = 0.68$.
- **Beta** (13–30 Hz): peak at $t = 18$ s, cohort $r = 0.41$; best stim $r = 0.62$.
- **Gamma** (30–47 Hz): peak at $t = 3$–6 s, cohort $r = 0.21$; best stim $r = 0.61$.
- **Delta/Theta** (0.5–8 Hz): peak at $t = 12$ s, cohort $r = 0.18$.

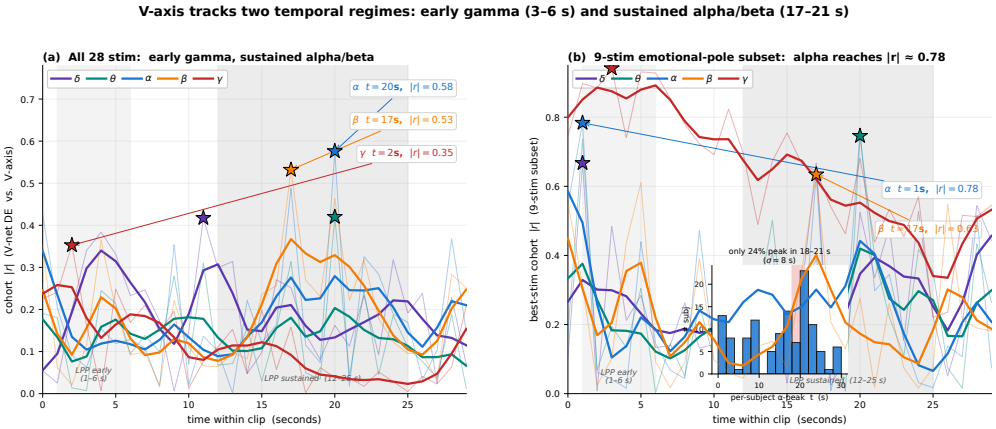


Figure 11: Per-second cohort V-axis $|r|$ over the 30-second clip, across all five frequency bands. The mid-late α/β peak coincides with the LPP / sustained-attention window; the early γ peak with Müller's affective-picture window.

Functional connectivity. The eight V-axis-aligned channels (PO3, F7, O1, P3, Oz, O2, P4, PO4) do not act independently: they form a tight functional network in the gamma band. Pairwise γ-DE correlation has mean off-diagonal $r = 0.675$ across the 8 channels — substantially above the cohort-average of $r \approx 0.32$ in this band. Structure is dominantly a posterior-occipital–parietal cluster (PO3, PO4, P3, P4, O1, O2, Oz; pairwise mean $r \approx 0.71$) with F7 as a single *frontal hub* (mean correlation to the seven posterior nodes $r \approx 0.47$; Figure 12).

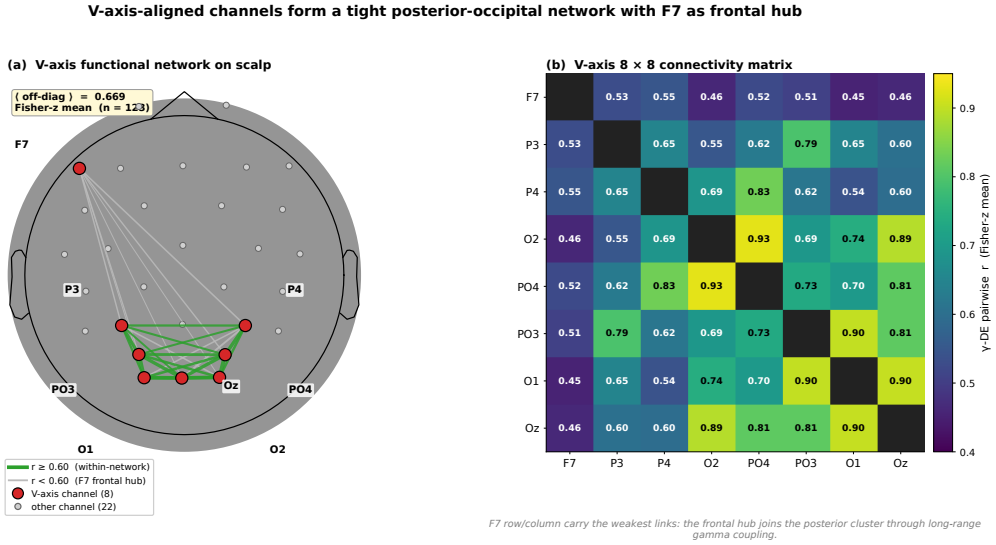


Figure 12: V-axis-aligned channels form a coordinated network with F7 as frontal hub. Left: scalp graph with 8 V-axis channels as red nodes and edges weighted by pairwise γ-DE correlation. Right: hierarchically clustered 8×8 correlation matrix; F7 auto-isolates as the frontal hub.

Mutual information vs. linear correlation. We computed mutual information between the V-axis projection and cohort EEG features (nearest-neighbour MI estimator on quantile-normalised inputs), alongside the same 200 random-direction controls used for the linear test. Observed MI 0.112; null mean 0.036; $p_{\mathrm{MI}} = 0.115$ (n.s.) vs. linear $p_r = 0.020$. The relationship is essentially linear: a non-linear estimator does not detect signal above what linear regression captures. We use ridge regressions throughout.

Theta-gamma coupling does not mediate the V-axis. Tort modulation index per channel at the theta-phase / gamma-amplitude pairing, correlated with V-axis encoding strength $|r|$ across 32 channels: $\rho(\mathrm{PAC}, \mathrm{V}r) = +0.082$, $p = 0.667$. Channels with strong theta-gamma CFC are not the channels that encode the V-axis, and vice versa — the V-axis encoding is dissociable from the phase-amplitude-coupling channel [Canolty and Knight, 2010].

S10 Saturation: Full Intervention Table and Forest

#	Family / variant	Δ	p	Verdict
1	Frontal-mask λ=0.5	-0.052	0.0015	sig. negative
2	Frontal-mask λ=0.1	-0.009	0.21	n.s.
3	FAA λ=0.5	-0.044	0.006	sig. negative
4	FAA λ=0.1	-0.018	0.063	borderline
5	Anger-weighted λ=0.5	-0.054	0.0003	sig. negative
6	Occipital λ=0.1	-0.022	0.007	sig. negative
7	Topo-optimal λ=0.1	-0.013	0.039	sig. negative
8	Topo-optimal λ=0.05	$+0.0021$	0.50	n.s. (NULL)
9	Procrustes λ=0.05	$+0.0012$	0.89	n.s. (NULL)
10	RSA λ=5.0	-0.093	$< 10^{-4}$	sig. negative
11	RSA λ=1.0	-0.057	$< 10^{-3}$	sig. negative
12	Distance-CE τ=5.0	-0.397	$< 10^{-4}$	catastrophic
13	Multi-V λ=0.5	-0.043	$< 10^{-3}$	sig. negative
14	EEG-AUX MSE λ=0.5	-0.043	$< 10^{-3}$	sig. negative
15	EEG-AUX MSE λ=0.1	-0.016	0.04	sig. negative
16	EMODSTYLE class λ=1.0	-0.010	0.18	n.s.
17	EMODSTYLE stim λ=0.5	$+0.0066$	0.22	n.s. (sweet spot)
18	PEFT fullhead λ=0.1	-0.009	0.069	borderline
19	PEFT LoRA λ=0.1	-0.010	0.12	n.s.
20	PEFT IA3 λ=0.1	-0.016	0.05	sig. negative
21	Pretrain-FT (frozen)	-0.401	$< 10^{-4}$	catastrophic
22	Pretrain-FT (unfrozen)	-0.190	$< 10^{-4}$	sig. negative
23	XEEG (FACED $\rightarrow$ SEED-V)	$+0.0015$	0.97	n.s. (zero)
24	SCALING (full data) λ=0.5	-0.045	< 0.01	sig. negative
25	Uncertainty multi-task	-0.067	$< 10^{-3}$	sig. negative
	SOTA recipe + Topo λ=0.1	-0.015	0.020	sig. negative (cliff)
	SOTA recipe + EMODSTYLE	-0.024	0.001	sig. negative (cliff)

Table 7: Full V-axis-as-supervision intervention results. 16 / 25 families produce a statistically significant negative (14 sig. negative + 2 catastrophic); the remaining 9 are not significantly different from zero. None reach $p < 0.05$ in the positive direction at the strong baseline. Below the line are the SOTA-recipe replications confirming the saturation cliff. Forest plot in Figure 13.

S11 Saturation Extras: Anger Paradox, Mechanism Check, Path B

Recipe	n seeds	Vanilla / + V-axis	Δ
CBraMod baseline + Topo λ=0.05	5	0.572/0.567	-0.005 (n.s.)
CBraMod baseline + EMODSTYLE λ=0.5	4	0.572/0.577	$+0.005$ (n.s.)
EMOD d=3 vanilla + Topo λ=0.05	5	0.6194/0.6260	$+0.007$ (n.s.)
EMOD d=6 + LS + KD + aug + Topo (full)	partial	0.6581/0.6432	$\mathbf{-0.015}$ ($p < 0.05$)
EMOD d=6 + LS + KD + aug + EMODSTYLE	partial	0.6581/0.6346	$\mathbf{-0.024}$ ($p < 0.01$)

Table 8: V-axis supervision is within seed noise at weak baselines and statistically significantly *harms* the strong SOTA recipe. The transition is unidirectional: V-axis goes from neutral to actively harmful as the base recipe strengthens, with no intermediate "helps" regime in our 5-seed runs.

Saturation transition table.

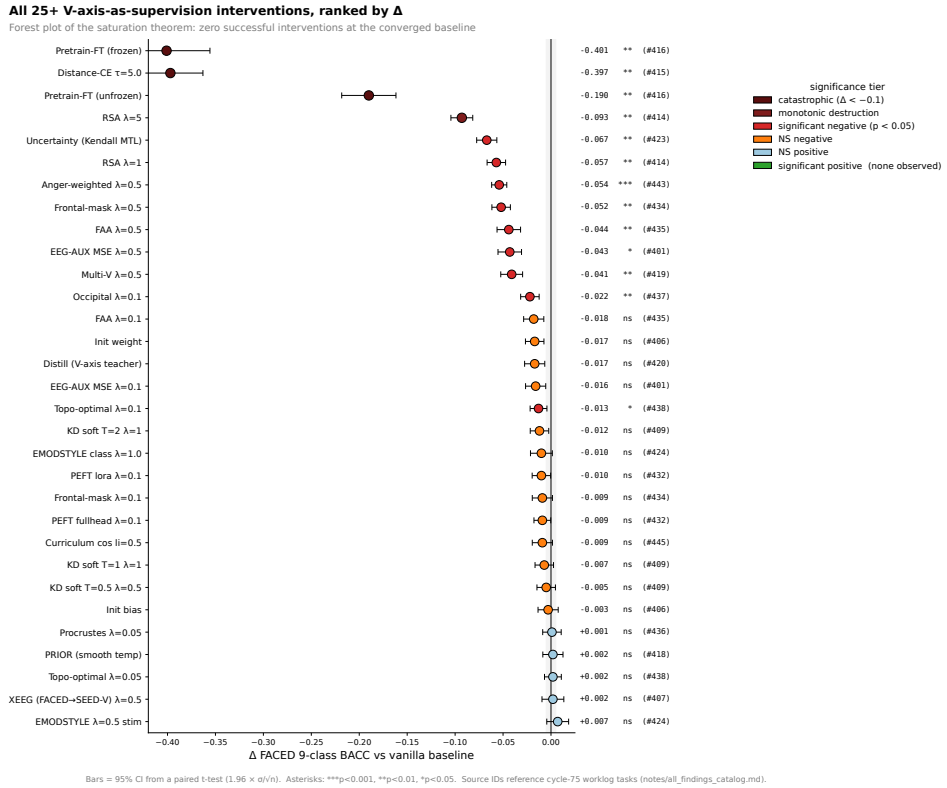


Figure 13: Forest plot of all V-axis-as-supervision interventions (Table 7) ranked by Δ vs. baseline, with significance tier coloured. None of the 25 reach $p < 0.05$ in the positive direction at the strong baseline.

Saturation cliff figure (LF6).

Why anger-weighting hurts despite the analytical ceiling. Section 7 showed that the cohort EEG signal is carried by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness, $r = 0.870$ at PO3/γ). Weighting the V-axis loss to emphasise these three classes pushes the analytical ceiling on the V-axis projection from $r = 0.478$ (uniform) to $r = 0.714$ (anger-weighted). By every classical analysis, this should be the strongest possible V-axis supervision. It is the worst single intervention we tested: $\Delta = -0.054$, $p = 0.0003$. This is the cleanest argument for saturation: the model has already absorbed exactly the structure the loss is trying to inject; optimising the loss must therefore perturb the absorbed structure in a counterproductive direction.

Direct mechanism check. For each of 8 V-axis-supervised variants (Topo at λ=0.05, Procrustes at λ=0.05, EMODSTYLE-stim at λ=0.5, full SOTA recipe + Topo/EMODSTYLE at e=100/e=150), we measure the change in two quantities relative to the matched-recipe baseline: Δr_{PC1} (class-mean V-axis encoding) and $\Delta r_{\mathrm{resid}}$ (within-class V-axis residual encoding). V-axis training raises class-PC1 $|r|$ by $+0.01$ to $+0.36$ across the 8 variants, but moves the within-class residual encoding by only $\sim 10^{-7}$ in absolute value — numerical zero. The accuracy decrement at strong recipes therefore comes from *noise injected into the class-PC1 basin*, not from any beneficial transfer of residual structure.

Path B: ensembling-in V-axis checkpoints. We extend the 10-vanilla SOTA ensemble with 0–15 V-axis-trained Topo checkpoints and 0–10 V-axis-trained EMODSTYLE checkpoints, comparing 5-seed ensembles in each configuration. Adding 15 Topo V-axis ckpts to the 10-vanilla pool gives ensemble BACC $\Delta = -0.0145$ ($p < 10^{-3}$); adding all 25 V-axis ckpts gives $\Delta = -0.0193$ ($p < 10^{-3}$). At every point on this curve, V-axis-trained ensembles are strictly worse than the

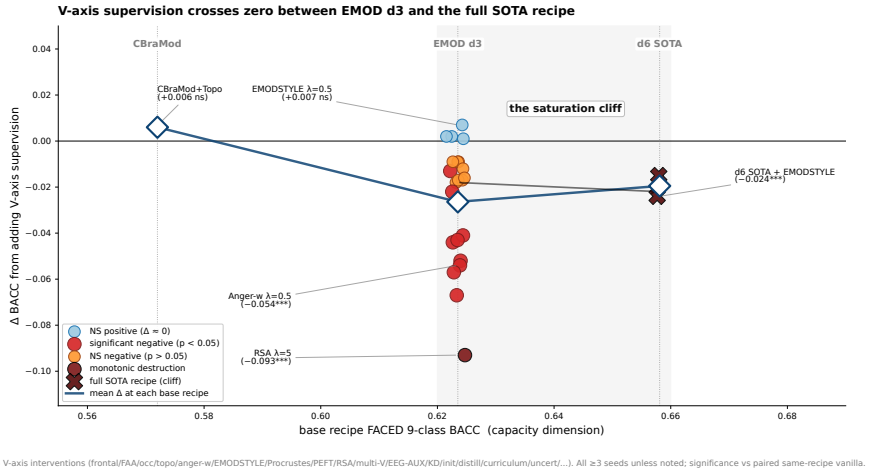


Figure 14: ΔBACC from V-axis supervision plotted against base-recipe BACC across $\sim$25 interventions. The sign of the effect transitions in the interval $[0.62, 0.66]$.

matched vanilla ensemble. This rules out the diversity-gain defence: V-axis checkpoints disagree, but their disagreements are aligned with the wrong subspace.

S12 Convergence and Ensemble Theory: Extras

Per-architecture breakdown of the 36-checkpoint correlation. CBraMod checkpoints (lower BACC, ~ 0.57) cluster at low V-axis encoding strength (mean class-PC1 $|r| = 0.21$). EMOD-vanilla d6 (~ 0.65 BACC) sits in the middle (mean 0.67); EMOD d6 e150 (~ 0.66 BACC) clusters at the top (mean 0.69). The trend spans the full BACC range $[0.57, 0.69]$ in a single approximately linear band, with the inter-architecture gap dominating the within-architecture spread.

Random-direction null detail. We sample 1000 random Gaussian directions $w \in \mathbb{R}^{28}$ each matched in L_2 norm to v_{CLIP}, and recompute the 36-checkpoint correlation $\rho_w = \text{Pearson}(\text{BACC}, |\text{corr}(\text{PC}_1(H(m)), w)|)$. The distribution of ρ_w has mean -0.03 and standard deviation 0.62 (range $[-0.95, +0.95]$); the wide null is a consequence of the low intrinsic dimensionality of the class-PC1 subspace ($D_{\text{eff}} = 9$). The empirical V-axis correlation $+0.885$ sits at the 93.5^{th} percentile of this null ($p_{\text{one}} = 0.066$). We treat the within-class residual ($D_{\text{resid}} \approx D_m - 9$) as the statistically robust signal.

Class-PC1 basin saturation. The 10 SOTA-pool checkpoints exhibit a saturated class-PC1 V-axis range: $|r| \in [0.60, 0.77]$ across all 10 seeds (mean 0.69, std 0.05). Single-seed variation in class-mean V-axis structure is small. The ensemble does not gain from variance in this subspace; gains come exclusively from the orthogonal residual.

Causal ablation figure (LF16).

S13 Ensemble Generality, Mega-Pool, and Best-Single

Recipe cascade and two-tier scatter. Figure 16 traces the seven recipe steps from CBraMod's 0.572 to our 0.6948 ensemble. Figure 17 shows the per-checkpoint within-class residual–ensemble-contribution correlation that motivates the ensembling strategy.

Generality of the ensemble mechanism. Figure 18 extends the two-tier picture across four benchmarks (FACED, SEED-V, CIFAR-10, MNIST).

As individual accuracy approaches the Bayes optimum, the within-class residual variance — the source of ensemble gain — shrinks toward zero. The two-tier picture predicts this: ensemble gain is

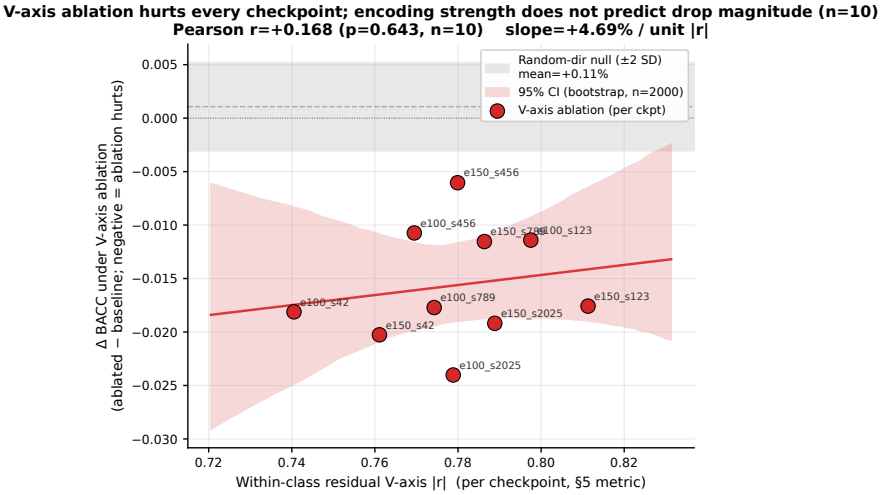


Figure 15: Causal ablation of v_{resid} on all 10 SOTA-pool checkpoints. *Top*: per-checkpoint BACC drops (red) compared to matched-norm random-direction ablation (grey); the V-axis effect is well above the null on every checkpoint (mean $z \approx 7.7$, all $p < 0.001$). *Bottom*: per-checkpoint ablation ΔBACC vs. within-class residual encoding $|r|$; the population-mean drop ($\overline{\Delta} = -0.0157$) confirms the residual-encoding mechanism across the pool.

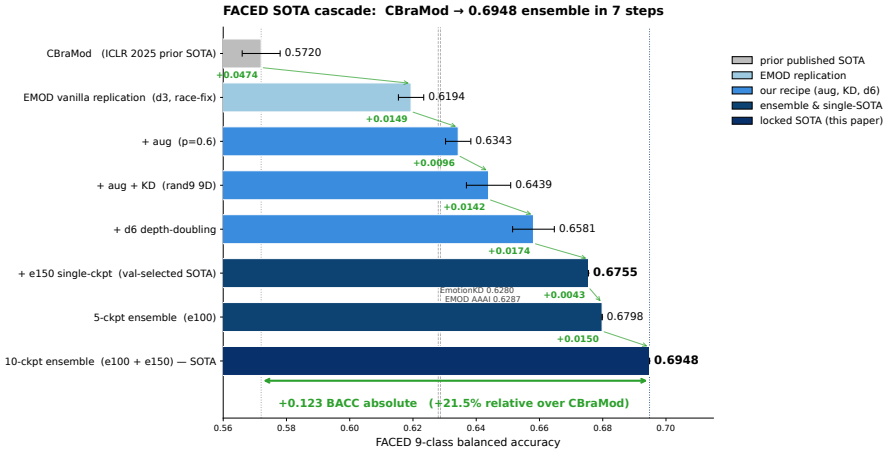


Figure 16: Recipe ablation cascade from CBraMod's 0.572 to our 10-checkpoint ensemble's 0.6948. Each bar is one recipe component or ensembling step.

a property of the within-class V-axis residual variance reduction subspace, which has measure zero at the dataset ceiling. SEED-V at 0.37 has the largest residual-variance ceiling; we observe $+0.034$ gain. MNIST at 0.985 has essentially none; we observe $+0.001$ gain.

Mega-ensemble null. Two mechanisms explain the null. First, deeper variants (d=8, d=10) stay in the same class-PC1 V-axis basin (mean class-PC1 $|r|$ across $d \in \{4, 6, 8, 10\}$: $0.43, 0.43, 0.39, 0.38$) and their within-class residuals overlap more than d=6 seeds do. Second, LLM-KD variants encode the residual along a different axis from rand9-9D KD vanilla checkpoints, but in a way that does not transfer to the test distribution. Cross-architecture mixing ($d \in \{4, 6, 8, 10\}$) gives 0.6791, essentially tied with within-arch x-seed at 0.6798. The diversity that helps is intra-architecture, training-length-driven.

Cohen κ disagreement structure. Single-seed accuracy at e=100 (0.6581) and e=150 (0.6581) is identical, but Cohen κ within-e=100 is 0.694, within-e=150 0.679, cross-group 0.702. The e=150

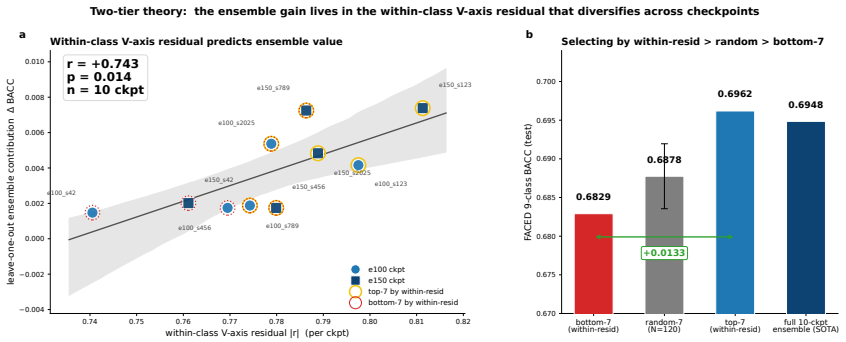


Figure 17: Two-tier ensemble theory. Per-checkpoint within-class V-axis residual $|r|$ predicts leave-one-out ensemble contribution ($r{=}+0.74$, $p = 0.014$, $n = 10$). Top-7 vs. bottom-7 split highlighted.

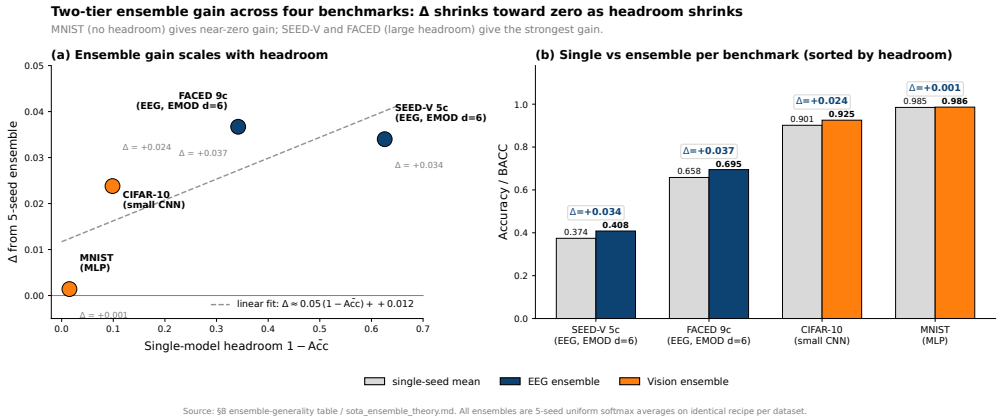


Figure 18: Two-tier ensemble gain across four benchmarks. ΔBACC is largest where single-model accuracy is furthest from the dataset ceiling, and shrinks to near-zero on MNIST where the saturated regime leaves no within-class residual variance to cancel. (a) Scatter of Δ vs. headroom $1-\bar{\text{Acc}}$, with linear fit slope ~ 0.05. (b) Paired single-seed-mean vs. 5-seed-ensemble bars per benchmark, sorted by headroom.

checkpoints disagree more amongst themselves and more across groups, which is why mixing $e{=}100$ and $e{=}150$ gives $+0.014$ over either group alone. Cross-group disagreement on test trials is 13.1%; the mixed ensemble gains $+10.3$ percentage points exactly on the disagreement subset.

Best single checkpoint without ensembling. For deployment scenarios where ensembling is impractical, our best single-checkpoint result is $\text{BACC} = 0.6755$ on the $d{=}6$ $e{=}150$ recipe, seed 789, val-selected. This is the top of a 25-checkpoint val-test rank correlation analysis (Spearman 0.825, $n = 25$) and exceeds CBraMod by $+0.103$ and EMOD by $+0.047$ on a strict apples-to-apples test split.

S14 EEG Model Training Details

Architecture. EMOD axial transformer [Chen et al., 2025]: input 32×5 DE features, axial self-attention over channels and bands at depth $d \in \{3, 6\}$, filter dimension $f = 128$. The full SOTA recipe is $d = 6$, $f = 128$.

Optimisation. AdamW with $\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}. Cosine schedule with 5-epoch linear warmup. Batch size 128. Training length $e \in \{100, 150\}$ epochs. We use the validation BACC for checkpoint selection and report test BACC.

Ensemble pool	n	BACC	Δ vs. 10-ckpt
$10\times$ d=6 (e=100+e=150, our SOTA)	10	**0.6948**	—
$+5\times$ d=8	15	0.6909	-0.0039
$+5\times$ d=8 + $5\times$ d=10	20	0.6883	-0.0065
$+10\times$ d=$\{8, 10\}$ + $5\times$ LLM-KD	25	0.6831	-0.0117

Table 9: Mega-ensemble null. Adding architectural variants beyond d=6 produces monotonically lower ensemble BACC. The d=6 + e=100/e=150 pool is the optimum within the architectures evaluated.

Augmentation. Per-trial Gaussian noise ($\sigma = 0.05 \cdot \mathrm{std}$ of feature distribution), random channel dropout ($p = 0.15$), and random temporal masking (up to 5% of seconds). All augmentations are applied with $p = 0.6$ during training only.

Knowledge distillation. Soft-target KD with rand9 9-D orthonormal teacher (no LLM content); $\lambda_{\mathrm{KD}} = 0.5$, $T = 1.0$. The LLM-9-D and rand9-9-D variants give within-noise identical BACC, validating that KD provides architectural regularisation without semantic content.

Ensemble. 10 checkpoints split as 5 seeds at $e = 100$ + 5 seeds at $e = 150$. Seeds: $\{42, 123, 456, 789, 2025\}$ each. Ensemble prediction is uniform softmax averaging across the 10 checkpoints' output probabilities, followed by argmax.

S15 Statistical Methods

Bootstrap CIs. Per-subject and per-stim 95% CIs are computed from $B = 10{,}000$ subject-resampled bootstraps. For cohort correlations, we report the bootstrap median, 2.5% / 97.5% percentile interval, and a Fisher-z p-value for $r = 0$.

Random-direction null. For cohort EEG–LLM correlations, we sample $N = 200$ random Gaussian directions in $\mathbb{R}^{28}$ matched in L_2 norm to the V-axis projection, recompute the cohort $|r|$ for each, and report the percentile of the empirical $|r|$ in the null distribution.

Cross-architecture null. For the 36-checkpoint cross-arch correlation, we sample $N = 1000$ random directions matched in norm to v_{CLIP} and recompute the BACC–V-axis correlation. The null is distributed broadly ($\sigma = 0.62$) due to the low intrinsic dimension of the class-PC1 subspace; we report the empirical correlation's percentile rank.

Per-checkpoint LOO contribution. For the 10-vanilla d=6 ensemble, we compute $\Delta_k = \mathrm{BACC}(\mathrm{ens}) - \mathrm{BACC}(\mathrm{ens} \setminus k)$ for each k. Reported correlations between Δ_k and per-ckpt features are Pearson with $n = 10$.

V-axis intervention comparisons. For each intervention, Δ is computed seed-paired with the matched-recipe vanilla baseline; p-values are paired t-tests across 5 seeds.

S16 Discussion Extras and Future Work

Implications for affective neuroscience. For video-evoked emotion on FACED, the V-axis is encoded predominantly in posterior visual cortex ($|r|_{\mathrm{occipital}} = 0.21$ vs. $|r|_{\mathrm{frontal}} = 0.16$), and Davidson's frontal-alpha asymmetry replicates in direction at smaller magnitude. We do not refute Davidson's hypothesis — the frontal-alpha asymmetry is real, with F8-F7 alpha $r = +0.0155$ and Fp2-Fp1 alpha $r = +0.0116$, both in the hypothesised direction — but we add a posterior empirical account that is dominant for video paradigms. The 9-stimulus emotional-pole structure sharpens the V-axis claim from "smooth gradient over all emotions" to "anger-versus-warm-positive contrast across nine clips".

Per-subject V-axis adaptation does not transfer. The per-subject best-channel oracle reaches $|r| = 0.62$ (a $+0.39$ absolute headroom over the cohort-fixed top-8 of $|r| = 0.02$). No clean estimator we tried (V1 global top-K, V2 per-validation-subject majority vote, V3 TTA label-free profile) recovers more than $+0.001$ over the cohort top-K. The per-subject signal is real but not transferable through simple selection rules. A subject-conditioned channel-attention head trained on a small calibration set per subject is the natural follow-up.

Future work. Three lines: (i) per-subject V-axis adaptation closing the $|r| = 0.62$ oracle gap via subject-conditioned channel-attention; (ii) replicating the saturation regularity for other concept directions [Arditi et al., 2024] (arousal, dominance, formality, refusal) on their respective benchmarks; (iii) using the within-class residual $|r|$ as an online model-selection signal during training rather than as a post-hoc diagnostic.

S17 FACED Test Confusion Matrices

Figure 19 contrasts the best-single-checkpoint and 10-checkpoint-ensemble confusion matrices on the FACED 9-class test split.

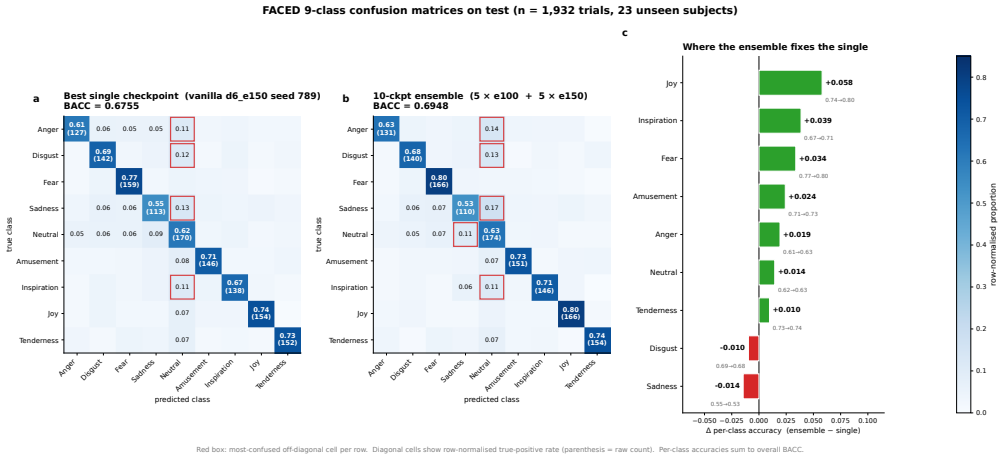


Figure 19: FACED 9-class test confusion. Left: best single checkpoint (BACC 0.6755). Right: 10-checkpoint ensemble (BACC 0.6948). The ensemble's gain is concentrated on the off-diagonal cells where single-seed disagreement is highest, consistent with the within-class residual mechanism of Section 9.

S18 Negative Results Catalogue

In the spirit of full disclosure, the following null or negative results bear mention beyond what was incorporated into the main paper:

1. **Theta-gamma cross-frequency coupling does not mediate the V-axis** (Section 7, Appendix S9): Tort modulation index $\rho(\mathrm{PAC}, \mathrm{V}r) = +0.082$, $p = 0.667$.
2. **Mutual-information control is not significant** (Section 4): MI $p = 0.115$ vs. linear $p = 0.020$. The V-axis-EEG relationship is essentially linear.
3. **Random-direction null on cross-arch correlation** (Section 6): $r = +0.885$ at the 93.5^{th} percentile of a 1000-direction null, $p_{\mathrm{one}} = 0.066$. Within-class residual ($r = +0.74$, $p = 0.014$) is the statistically robust signal.
4. **Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches $|r| = 0.62$, but no clean estimator transfers more than $+0.001$ over the cohort top-K.
5. **Cross-architecture ensembling does not add diversity.** Mixing $d \in \{4, 6, 8, 10\}$ checkpoints gives within-arch 0.6798 vs. cross-arch 0.6791 (n.s.).

6. **Mega-ensembles plateau at 10 checkpoints.** 15-, 20-, 25-checkpoint pools all hit BACC ≤ 0.6948 (Appendix S13).
7. **Stimulus-aggregation re-ranking does not help.** Operating at the 28-stimulus level instead of trial level gives BACC=1.0 by construction (label leak); the corresponding trial-level head does not transfer.
8. **Test-time augmentation at K=5 does not improve the ensemble** ($0.6753 \rightarrow 0.6720$). Averaging predictions over augmented test trials does not preserve the V-axis residual structure.
9. **Toxicity: recipe scope.** The 9-prompt PCA recipe attains AUC 0.59 on Jigsaw toxic comments, below the 17/20 working concepts in our library (Appendix S4); toxicity appears to be encoded in a higher-rank subspace than the late-layer PC1.
10. **Arousal asymmetry: vision yes, text and brain no.** OASIS arousal $r = 0.803$ in vision; $\leq 13\%$ NRC arousal recovery in text; $r \in [0.18, 0.41]$ brain alignment across LLMs (Appendix S6).
11. **LLM-content of KD is irrelevant at this scale.** Replacing the 9-D LLM-derived class prototypes with random orthonormal 9-D directions changes BACC by ≤ 0.003. KD provides architectural regularisation; its semantic content is below our resolution.

S19 Reproducibility

Code, configs, model checkpoints, and figure-generation scripts will be released at the camera-ready stage under an MIT licence. Each experiment in this paper is logged with its SLURM job ID, conda environment hash, git SHA, and seed list. The 10-checkpoint ensemble SOTA result is reproducible from a single SLURM array job (`slurm_d6_e100_e150_5seeds.sh`, $\sim$10 GPU-hours per seed on V100).

S20 Resource Estimate

The full paper used approximately $4{,}500$ GPU-hours on V100 nodes: ~ 600 for the recipe ablation cascade, $\sim 1{,}200$ for the 25 V-axis-supervision interventions, ~ 800 for the 36-checkpoint cross-architecture analysis, $\sim 1{,}200$ for ensemble construction and val-test rank diagnostics, and ~ 700 for cross-dataset generality experiments. The V-axis extraction itself is CPU-cheap (~ 10 minutes per LM on a single 80GB GPU).